

The most important concept in elementary calculus is that of the limit. Recall that $\lim_{x \rightarrow x_0} f(x) = L$ intuitively means that values $f(x)$ of the function f can be made arbitrarily close to the real number L if values of x are chosen sufficiently close to, but not equal to, the real number x_0 . In real analysis, the concepts of continuity, the derivative, and the definite integral were *all* defined using the concept of a limit. Complex limits play an equally important role in study of complex analysis. The concept of a complex limit is similar to that of a real limit in the sense that $\lim_{z \rightarrow z_0} f(z) = L$ will mean that the values $f(z)$ of the complex function f can be made arbitrarily close the complex number L if values of z are chosen sufficiently close to, but not equal to, the complex number z_0 . Although outwardly similar, there is an important difference between these two concepts of limit. In a real limit, there are two directions from which x can approach x_0 on the real line, namely, from the left or from the right. In a complex limit, however, there are infinitely many directions from which z can approach z_0 in the complex plane. In order for a complex limit to exist, each way in which z can approach z_0 must yield the same limiting value.

In this section we will define the limit of a complex function, examine some of its properties, and introduce the concept of continuity for functions of a complex variable.

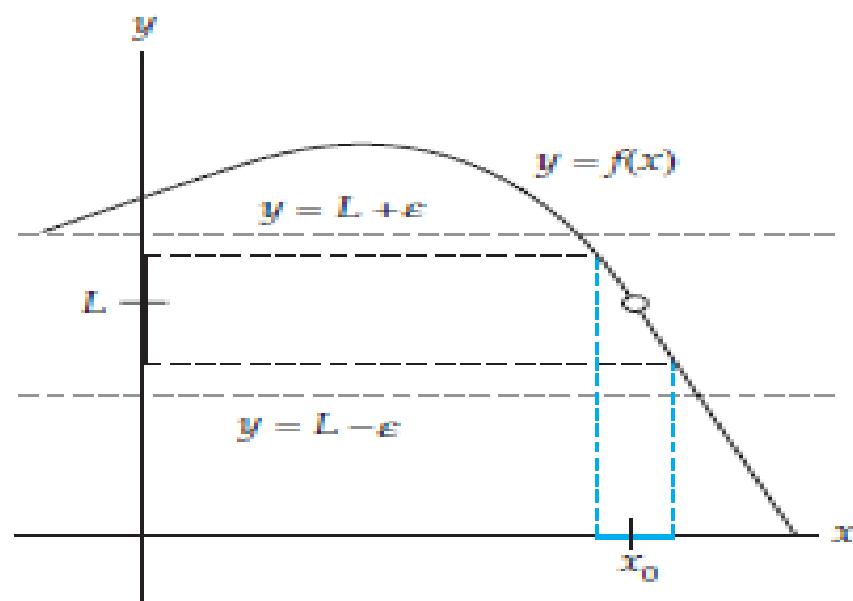
Limits and Continuity

Limit of a Real Function $f(x)$

The limit of f as x tends x_0 exists and is equal to L

if for every $\varepsilon > 0$ there exists a $\delta > 0$ such that $|f(x) - L| < \varepsilon$ (1)

whenever $0 < |x - x_0| < \delta$.

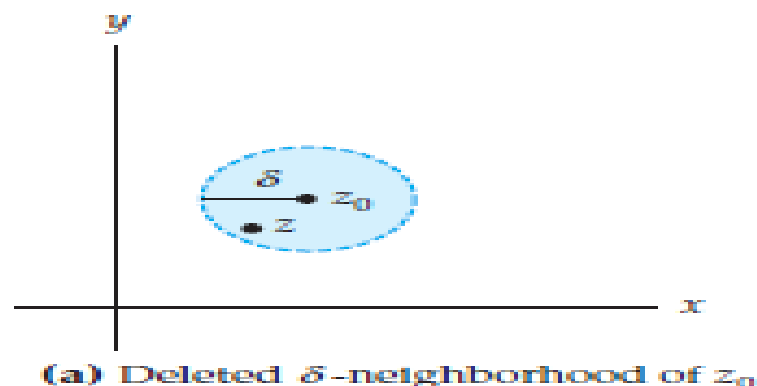


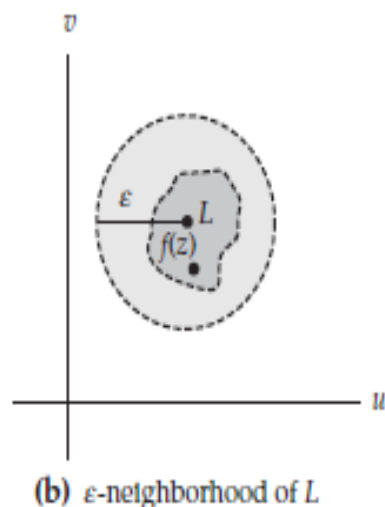
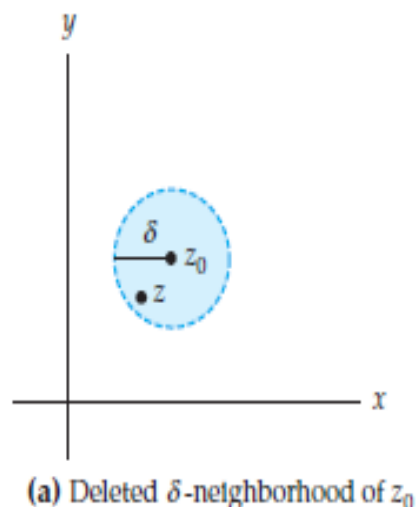
Complex Limits

A complex limit is, in essence, the same as a real limit except that it is based on a notion of “close” in the complex plane. Because the distance in the complex plane between two points z_1 and z_2 is given by the modulus of the difference of z_1 and z_2 , the precise definition of a complex limit will involve $|z_2 - z_1|$. For example, the phrase “ $f(z)$ can be made arbitrarily close to the complex number L ,” can be stated precisely as: for every $\varepsilon > 0$, z can be chosen so that $|f(z) - L| < \varepsilon$. Since the modulus of a complex number is a *real* number, both ε and δ still represent small positive *real* numbers in the following definition of a complex limit. The complex analogue of (1) is:

Definition Limit of a Complex Function

Suppose that a complex function f is defined in a deleted neighborhood of z_0 and suppose that L is a complex number. The limit of f as z tends to z_0 exists and is equal to L , written as $\lim_{z \rightarrow z_0} f(z) = L$, if for every $\varepsilon > 0$ there exists a $\delta > 0$ such that $|f(z) - L| < \varepsilon$ whenever $0 < |z - z_0| < \delta$.

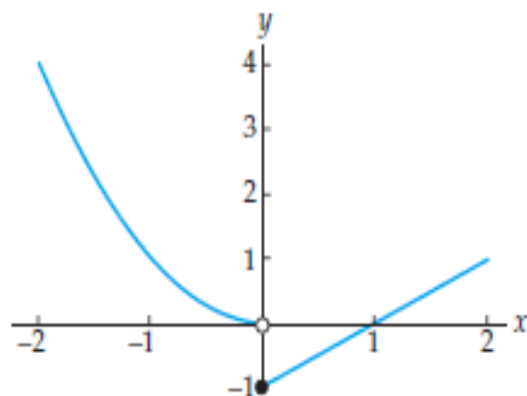




Because a complex function f has no graph, we rely on the concept of complex mappings to gain a geometric understanding of Definition 1. Recall that the set of points w in the complex plane satisfying $|w - L| < \varepsilon$ is called a neighborhood of L , and that this set consists of all points in the complex plane lying within, but not on, a circle of radius ε centered at the point L . Also recall that the set of points satisfying the inequalities $0 < |z - z_0| < \delta$ is called a deleted neighborhood of z_0 and consists of all points in the neighborhood $|z - z_0| < \delta$ excluding the point z_0 . By Definition 1, if $\lim_{z \rightarrow z_0} f(z) = L$ and if ε is any positive number, then there is a deleted neighborhood of z_0 of radius δ with the property that for every point z in this deleted neighborhood, $f(z)$ is in the ε neighborhood of L . That is, f maps the deleted neighborhood $0 < |z - z_0| < \delta$ in the z -plane into the neighborhood $|w - L| < \varepsilon$ in the w -plane. the deleted neighborhood of z_0 shown in color is mapped onto the set shown in dark gray in Figure 1(b). As required by Definition 1, the image lies within the ε -neighborhood of L shown in light gray in Figure 1(b).

Complex and real limits have many common properties, but there is at least one very important difference. For real functions, $\lim_{x \rightarrow x_0} f(x) = L$ if and only if $\lim_{x \rightarrow x_0^+} f(x) = L$ and $\lim_{x \rightarrow x_0^-} f(x) = L$. That is, there are two directions from which x can approach x_0 on the real line, from the right (denoted by $x \rightarrow x_0^+$) or from the left (denoted by $x \rightarrow x_0^-$). The real limit exists if and only if these two one-sided limits have the same value. For example, consider the real function defined by:

$$f(x) = \begin{cases} x^2, & x < 0 \\ x - 1, & x \geq 0 \end{cases}.$$



The limit of f does not exist as x approaches 0.

The limit of f as x approaches 0 does not exist since $\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} x^2 = 0$, but $\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} (x - 1) = -1$.

For limits of complex functions, z is allowed to approach z_0 from *any* direction in the complex plane, that is, along any curve or path through z_0 .

In order that $\lim_{z \rightarrow z_0} f(z)$ exists and equals L , we require that $f(z)$ approach the same complex number L along every possible curve through z_0 .

Criterion for the Nonexistence of a Limit

If f approaches two complex numbers $L_1 \neq L_2$ for two different curves or paths through z_0 , then $\lim_{z \rightarrow z_0} f(z)$ does not exist.

EXAMPLE 1 A Limit That Does Not Exist

Show that $\lim_{z \rightarrow 0} \frac{z}{\bar{z}}$ does not exist.

Solution We show that this limit does not exist by finding two different ways of letting z approach 0 that yield different values for $\lim_{z \rightarrow 0} \frac{z}{\bar{z}}$. First, we let z approach 0 along the real axis. That is, we consider complex numbers of the form $z = x + 0i$ where the real number x is approaching 0. For these points we have:

$$\lim_{z \rightarrow 0} \frac{z}{\bar{z}} = \lim_{x \rightarrow 0} \frac{x + 0i}{x - 0i} = \lim_{x \rightarrow 0} 1 = 1. \quad (2)$$

On the other hand, if we let z approach 0 along the imaginary axis, then $z = 0 + iy$ where the real number y is approaching 0. For this approach we have:

$$\lim_{z \rightarrow 0} \frac{z}{\bar{z}} = \lim_{y \rightarrow 0} \frac{0 + iy}{0 - iy} = \lim_{y \rightarrow 0} (-1) = -1. \quad (3)$$

Since the values in (2) and (3) are not the same, we conclude that $\lim_{z \rightarrow 0} \frac{z}{\bar{z}}$ does not exist. Limit must be independent of direction approach

The limit $\lim_{z \rightarrow 0} \frac{z}{\bar{z}}$ from Example 1 did not exist because the values of $\lim_{z \rightarrow 0} \frac{z}{\bar{z}}$ as z approached 0 along the real and imaginary axes did not agree. However, even if these two values did agree, the complex limit may still fail to exist.

In general, computing values of $\lim_{z \rightarrow z_0} f(z)$ as z approaches z_0 from different directions, as in Example 1, can prove that a limit *does not exist*, but this technique cannot be used to prove that a limit *does exist*. In order to prove that a limit does exist we must use Definition 1.1.1 directly. This requires demonstrating that for every positive real number ε there is an appropriate choice of δ that meets the requirements of Definition 1.1.1. Such proofs are commonly called “epsilon-delta proofs.” Even for relatively simple functions, epsilon-delta proofs can be quite involved. Since this is an introductory text, we restrict our attention to what, in our opinion, are straightforward examples of epsilon-delta proofs.

EXAMPLE 2 An Epsilon-Delta Proof of a Limit

Prove that $\lim_{z \rightarrow 1+i} (2+i)z = 1+3i$.

Solution According to Definition $\lim_{z \rightarrow 1+i} (2+i)z = 1+3i$, if, for every $\varepsilon > 0$, there is a $\delta > 0$ such that $|(2+i)z - (1+3i)| < \varepsilon$ whenever $0 < |z - (1+i)| < \delta$. Proving that the limit exists requires that we find an appropriate value of δ for a given value of ε . In other words, for a given value of ε we must find a positive number δ with the property that if $0 < |z - (1+i)| < \delta$, then $|(2+i)z - (1+3i)| < \varepsilon$. One way of finding δ is to “work backwards.” The idea is to start with the inequality:

$$|(2+i)z - (1+3i)| < \varepsilon \quad (4)$$

and then use properties of complex numbers and the modulus to manipulate this inequality until it involves the expression $|z - (1+i)|$. Thus, a natural first step is to factor $(2+i)$ out of the left-hand side of (4):

$$|2+i| \cdot \left| z - \frac{1+3i}{2+i} \right| < \varepsilon. \quad (5)$$

Because $|2+i| = \sqrt{5}$ and $\frac{1+3i}{2+i} = 1+i$, (5) is equivalent to:

$$\sqrt{5} \cdot |z - (1+i)| < \varepsilon \quad \text{or} \quad |z - (1+i)| < \frac{\varepsilon}{\sqrt{5}}. \quad (6)$$

Thus, (6) indicates that we should take $\delta = \varepsilon/\sqrt{5}$. Keep in mind that the choice of δ is not unique. Our choice of $\delta = \varepsilon/\sqrt{5}$ is a result of the particular algebraic manipulations that we employed to obtain (6). Having found δ we now present the formal proof that $\lim_{z \rightarrow 1+i} (2+i)z = 1+3i$ that does not indicate how the choice of δ was made:

Given $\varepsilon > 0$, let $\delta = \varepsilon/\sqrt{5}$. If $0 < |z - (1 + i)| < \delta$, then we have $|z - (1 + i)| < \varepsilon/\sqrt{5}$. Multiplying both sides of the last inequality by $|2 + i| = \sqrt{5}$ we obtain:

$$|2 + i| \cdot |z - (1 + i)| < \sqrt{5} \cdot \frac{\varepsilon}{\sqrt{5}} \quad \text{or} \quad |(2 + i)z - (1 + 3i)| < \varepsilon.$$

Therefore, $|(2 + i)z - (1 + 3i)| < \varepsilon$ whenever $0 < |z - (1 + i)| < \delta$. So, according to Definition 1.1, we have proven that $\lim_{z \rightarrow 1+i} (2 + i)z = 1 + 3i$.

Limit of the Real Function $F(x, y)$

The limit of F as (x, y) tends to (x_0, y_0) exists and is equal to the real number L if for every $\varepsilon > 0$ there exists a $\delta > 0$ such that $|F(x, y) - L| < \varepsilon$ whenever $0 < \sqrt{(x - x_0)^2 + (y - y_0)^2} < \delta$. (7)

The expression $\sqrt{(x - x_0)^2 + (y - y_0)^2}$ in (7) represents the distance between the points (x, y) and (x_0, y_0) in the Cartesian plane. Using (7), it is relatively easy to prove that:

$$\lim_{(x,y) \rightarrow (x_0,y_0)} 1 = 1, \quad \lim_{(x,y) \rightarrow (x_0,y_0)} x = x_0, \quad \text{and} \quad \lim_{(x,y) \rightarrow (x_0,y_0)} y = y_0. \quad (8)$$

If $\lim_{(x,y) \rightarrow (x_0,y_0)} F(x, y) = L$ and $\lim_{(x,y) \rightarrow (x_0,y_0)} G(x, y) = M$, then (7) can also be used to show:

$$\lim_{(x,y) \rightarrow (x_0,y_0)} cF(x, y) = cL, \quad c \text{ a real constant}, \quad (9)$$

$$\lim_{(x,y) \rightarrow (x_0,y_0)} (F(x, y) \pm G(x, y)) = L \pm M, \quad (10)$$

$$\lim_{(x,y) \rightarrow (x_0,y_0)} F(x, y) \cdot G(x, y) = L \cdot M, \quad (11)$$

and
$$\lim_{(x,y) \rightarrow (x_0,y_0)} \frac{F(x, y)}{G(x, y)} = \frac{L}{M}, \quad M \neq 0. \quad (12)$$

Limits involving polynomial expressions in x and y can be easily computed using the limits in (8) combined with properties (9)–(12). For example,

$$\begin{aligned} \lim_{(x,y) \rightarrow (1,2)} (3xy^2 - y) &= 3 \left(\lim_{(x,y) \rightarrow (1,2)} x \right) \left(\lim_{(x,y) \rightarrow (1,2)} y \right) \left(\lim_{(x,y) \rightarrow (1,2)} y \right) - \lim_{(x,y) \rightarrow (1,2)} y \\ &= 3 \cdot 1 \cdot 2 \cdot 2 - 2 = 10. \end{aligned}$$

In general, if $p(x, y)$ is a two-variable polynomial function, then (8)–(12) can be used to show that

$$\lim_{(x,y) \rightarrow (x_0,y_0)} p(x, y) = p(x_0, y_0). \quad (13)$$

If $p(x, y)$ and $q(x, y)$ are two-variable polynomial functions and $q(x_0, y_0) \neq 0$, then (13) and (12) give:

$$\lim_{(x,y) \rightarrow (x_0,y_0)} \frac{p(x, y)}{q(x, y)} = \frac{p(x_0, y_0)}{q(x_0, y_0)}. \quad (14)$$

Theorem Real and Imaginary Parts of a Limit

Suppose that $f(z) = u(x, y) + iv(x, y)$, $z_0 = x_0 + iy_0$, and $L = u_0 + iv_0$. Then $\lim_{z \rightarrow z_0} f(z) = L$ if and only if

$$\lim_{(x,y) \rightarrow (x_0,y_0)} u(x, y) = u_0 \quad \text{and} \quad \lim_{(x,y) \rightarrow (x_0,y_0)} v(x, y) = v_0.$$

EXAMPLE 3 Using Theorem 13.1 to Compute a Limit

Use Theorem 13.1 to compute $\lim_{z \rightarrow 1+i} (z^2 + i)$.

Solution Since $f(z) = z^2 + i = x^2 - y^2 + (2xy + 1)i$, we can apply Theorem 13.1 with $u(x, y) = x^2 - y^2$, $v(x, y) = 2xy + 1$, and $z_0 = 1 + i$. Identifying $x_0 = 1$ and $y_0 = 1$, we find u_0 and v_0 by computing the two real limits:

$$u_0 = \lim_{(x,y) \rightarrow (1,1)} (x^2 - y^2) \quad \text{and} \quad v_0 = \lim_{(x,y) \rightarrow (1,1)} (2xy + 1).$$

Since both of these limits involve only multivariable polynomial functions, we can use (13) to obtain:

$$u_0 = \lim_{(x,y) \rightarrow (1,1)} (x^2 - y^2) = 1^2 - 1^2 = 0$$

and
$$v_0 = \lim_{(x,y) \rightarrow (1,1)} (2xy + 1) = 2 \cdot 1 \cdot 1 + 1 = 3,$$

and so $L = u_0 + iv_0 = 0 + i(3) = 3i$. Therefore, $\lim_{z \rightarrow 1+i} (z^2 + i) = 3i$.

Theorem Properties of Complex Limits

Suppose that f and g are complex functions. If $\lim_{z \rightarrow z_0} f(z) = L$ and $\lim_{z \rightarrow z_0} g(z) = M$, then

$$(i) \quad \lim_{z \rightarrow z_0} cf(z) = cL, \text{ } c \text{ a complex constant,}$$

$$(ii) \quad \lim_{z \rightarrow z_0} (f(z) \pm g(z)) = L \pm M,$$

$$(iii) \quad \lim_{z \rightarrow z_0} f(z) \cdot g(z) = L \cdot M, \text{ and}$$

$$(iv) \quad \lim_{z \rightarrow z_0} \frac{f(z)}{g(z)} = \frac{L}{M}, \text{ provided } M \neq 0.$$

$$\lim_{z \rightarrow z_0} c = c, \text{ } c \text{ a complex constant,} \tag{15}$$

$$\lim_{z \rightarrow z_0} z = z_0. \tag{16}$$

EXAMPLE 4 Computing Limits with Theorem

Use Theorem and the basic limits (15) and (16) to compute the limits

$$(a) \lim_{z \rightarrow i} \frac{(3+i)z^4 - z^2 + 2z}{z+1}$$

$$(b) \lim_{z \rightarrow 1+\sqrt{3}i} \frac{z^2 - 2z + 4}{z - 1 - \sqrt{3}i}$$

Solution

(a) By Theorem (iii) and (16), we have:

$$\lim_{z \rightarrow i} z^2 = \lim_{z \rightarrow i} z \cdot z = \left(\lim_{z \rightarrow i} z \right) \cdot \left(\lim_{z \rightarrow i} z \right) = i \cdot i = -1.$$

Similarly, $\lim_{z \rightarrow i} z^4 = i^4 = 1$. Using these limits, Theorems (i), (ii), and the limit in (16), we obtain:

$$\begin{aligned} \lim_{z \rightarrow i} ((3+i)z^4 - z^2 + 2z) &= (3+i) \lim_{z \rightarrow i} z^4 - \lim_{z \rightarrow i} z^2 + 2 \lim_{z \rightarrow i} z \\ &= (3+i)(1) - (-1) + 2(i) \\ &= 4 + 3i, \end{aligned}$$

and $\lim_{z \rightarrow i} (z+1) = 1+i$. Therefore, by Theorem (iv), we have:

$$\lim_{z \rightarrow i} \frac{(3+i)z^4 - z^2 + 2z}{z+1} = \frac{\lim_{z \rightarrow i} ((3+i)z^4 - z^2 + 2z)}{\lim_{z \rightarrow i} (z+1)} = \frac{4+3i}{1+i}.$$

After carrying out the division, we obtain $\lim_{z \rightarrow i} \frac{(3+i)z^4 - z^2 + 2z}{z+1} = \frac{7}{2} - \frac{1}{2}i$.

(b) In order to find $\lim_{z \rightarrow 1+\sqrt{3}i} \frac{z^2 - 2z + 4}{z - 1 - \sqrt{3}i}$, we proceed as in (a):

$$\begin{aligned} \lim_{z \rightarrow 1+\sqrt{3}i} (z^2 - 2z + 4) &= \left(1 + \sqrt{3}i\right)^2 - 2\left(1 + \sqrt{3}i\right) + 4 \\ &= -2 + 2\sqrt{3}i - 2 - 2\sqrt{3}i + 4 = 0, \end{aligned}$$

and $\lim_{z \rightarrow 1+\sqrt{3}i} (z - 1 - \sqrt{3}i) = 1 + \sqrt{3}i - 1 - \sqrt{3}i = 0$. It appears that

we cannot apply Theorem 1.4(iv) since the limit of the denominator is 0.

However, in the previous calculation we found that $1 + \sqrt{3}i$ is a root of the quadratic polynomial $z^2 - 2z + 4$. recall that if z_1 is

a root of a quadratic polynomial, then $z - z_1$ is a factor of the polynomial.

Using long division, we find that

$$z^2 - 2z + 4 = \left(z - 1 + \sqrt{3}i\right) \left(z - 1 - \sqrt{3}i\right).$$

$$\begin{aligned}\lim_{z \rightarrow 1 + \sqrt{3}i} \frac{z^2 - 2z + 4}{z - 1 - \sqrt{3}i} &= \lim_{z \rightarrow 1 + \sqrt{3}i} \frac{(z - 1 + \sqrt{3}i)(z - 1 - \sqrt{3}i)}{z - 1 - \sqrt{3}i} \\ &= \lim_{z \rightarrow 1 + \sqrt{3}i} (z - 1 + \sqrt{3}i).\end{aligned}$$

By Theorem (ii) and the limits in (15) and (16), we then have

$$\lim_{z \rightarrow 1 + \sqrt{3}i} (z - 1 + \sqrt{3}i) = 1 + \sqrt{3}i - 1 + \sqrt{3}i = 2\sqrt{3}i.$$

Therefore, $\lim_{z \rightarrow 1 + \sqrt{3}i} \frac{z^2 - 2z + 4}{z - 1 - \sqrt{3}i} = 2\sqrt{3}i.$

Continuity of Real Functions

Recall that if the limit of a real function f as x approaches the point x_0 exists and agrees with the value of the function f at x_0 , then we say that f is *continuous* at the point x_0 .

Continuity of a Real Function $f(x)$

A function f is continuous at a point x_0 if $\lim_{x \rightarrow x_0} f(x) = f(x_0)$. (17)

In real analysis, we visualize the concept of continuity using the graph of the function f . Informally, a function f is continuous if there are no breaks or holes in the graph of f . Since we cannot graph a complex function, our discussion of the continuity of complex functions will be primarily algebraic in nature.

Definition Continuity of a Complex Function

A complex function f is continuous at a point z_0 if

$$\lim_{z \rightarrow z_0} f(z) = f(z_0).$$

Criteria for Continuity at a Point

A complex function f is continuous at a point z_0 if each of the following three conditions hold:

- (i) $\lim_{z \rightarrow z_0} f(z)$ exists,*
- (ii) f is defined at z_0 , and*
- (iii) $\lim_{z \rightarrow z_0} f(z) = f(z_0)$.*

If a complex function f is not continuous at a point z_0 then we say that f is discontinuous at z_0 . For example, the function $f(z) = \frac{1}{1+z^2}$ is discontinuous at $z = i$ and $z = -i$.

EXAMPLE 5 Checking Continuity at a Point

Consider the function $f(z) = z^2 - iz + 2$. In order to determine if f is continuous at, say, the point $z_0 = 1 - i$, we must find $\lim_{z \rightarrow z_0} f(z)$ and $f(z_0)$, then check to see whether these two complex values are equal. From Theorem and the limits in (15) and (16) we obtain:

$$\lim_{z \rightarrow z_0} f(z) = \lim_{z \rightarrow 1-i} (z^2 - iz + 2) = (1-i)^2 - i(1-i) + 2 = 1 - 3i.$$

Furthermore, for $z_0 = 1 - i$ we have:

$$f(z_0) = f(1-i) = (1-i)^2 - i(1-i) + 2 = 1 - 3i.$$

Since $\lim_{z \rightarrow z_0} f(z) = f(z_0)$, we conclude that $f(z) = z^2 - iz + 2$ is continuous at the point $z_0 = 1 - i$.

Properties of Continuous Functions

Because the concept of continuity is defined using the complex limit, various properties of complex limits can be translated into statements about continuity.

Continuity of a Real Function $F(x, y)$

A function F is continuous at a point (x_0, y_0) if

$$\lim_{(x,y) \rightarrow (x_0,y_0)} F(x, y) = F(x_0, y_0).$$

Theorem Real and Imaginary Parts of a Continuous Function

Suppose that $f(z) = u(x, y) + iv(x, y)$ and $z_0 = x_0 + iy_0$. Then the complex function f is continuous at the point z_0 if and only if both real functions u and v are continuous at the point (x_0, y_0) .

EXAMPLE 7 Checking Continuity Using Theorem

Show that the function $f(z) = \bar{z}$ is continuous on $\mathbb{C}$.

Solution According to Theorem 14.1, $f(z) = \bar{z} = \overline{x + iy} = x - iy$ is continuous at $z_0 = x_0 + iy_0$ if both $u(x, y) = x$ and $v(x, y) = -y$ are continuous at (x_0, y_0) . Because u and v are two-variable polynomial functions, it follows from (13) that:

$$\lim_{(x,y) \rightarrow (x_0,y_0)} u(x, y) = x_0 \quad \text{and} \quad \lim_{(x,y) \rightarrow (x_0,y_0)} v(x, y) = -y_0.$$

This implies that u and v are continuous at (x_0, y_0) , and, therefore, that f is continuous at $z_0 = x_0 + iy_0$ by Theorem 14.1. Since $z_0 = x_0 + iy_0$ was an arbitrary point, we conclude that the function $f(z) = \bar{z}$ is continuous on $\mathbb{C}$.

Theorem Properties of Continuous Functions

If f and g are continuous at the point z_0 , then the following functions are continuous at the point z_0 :

- (i) cf , c a complex constant,
- (ii) $f \pm g$,
- (iii) $f \cdot g$, and
- (iv) $\frac{f}{g}$ provided $g(z_0) \neq 0$.

Polynomial functions are continuous on the entire complex plane $\mathbb{C}$.

Continuity of Rational Functions

Rational functions are continuous on their domains.

- (i) Analogous to real analysis, we can also define the concepts of infinite limits and limits at infinity for complex functions. Intuitively, the limit $\lim_{z \rightarrow \infty} f(z) = L$ means that values $f(z)$ of the function f can be made arbitrarily close to L if values of z are chosen so that $|z|$ is sufficiently large. A precise statement of a limit at infinity is:

The limit of f as z tends to ∞ exists and is equal to L if for every $\varepsilon > 0$ there exists a $\delta > 0$ such that $|f(z) - L| < \varepsilon$ whenever $|z| > 1/\delta$.

Using this definition it is not hard to show that:

$$\lim_{z \rightarrow \infty} f(z) = L \text{ if and only if } \lim_{z \rightarrow 0} f\left(\frac{1}{z}\right) = L.$$

Similarly, the infinite limit $\lim_{z \rightarrow z_0} f(z) = \infty$ is defined by:

The limit of f as z tends to z_0 is ∞ if for every $\varepsilon > 0$ there is a $\delta > 0$ such that $|f(z)| > 1/\varepsilon$ whenever $0 < |z - z_0| < \delta$.

From this definition we obtain the following result:

$$\lim_{z \rightarrow z_0} f(z) = \infty \text{ if and only if } \lim_{z \rightarrow z_0} \frac{1}{f(z)} = 0.$$

Show that the following limits are true

$$\lim_{z \rightarrow z_0} (az + c) = az_0 + c$$

$$|f(z) - (az_0 + c)| < \epsilon$$

$$|az + c - az_0 - c| = |a| |z - z_0| < \epsilon$$

$$|z - z_0| < \frac{\epsilon}{|a|}$$

$\delta = \frac{\epsilon}{|a|}$; so, the required is obtained

$$\lim_{z \rightarrow z_0} (z^2 + c) = z_0^2 + c$$

$$|f(z) - (z_0^2 + c)| = |z^2 + c - z_0^2 - c| = |z^2 - z_0^2|$$

$$|z^2 - z_0^2| = |z - z_0| |z + z_0|$$

Let $|z - z_0| < 1$;

if so, $|z + z_0| = |z - z_0 + 2z_0| \leq |z - z_0| + |2z_0| < 1 + 2|z_0|$

$$|f(z) - w_0| = |z - z_0| |z + z_0| < |z - z_0| (1 + 2|z_0|) < \varepsilon$$

$$|z - z_0| < \frac{\varepsilon}{1 + 2|z_0|}$$

if $\delta = \min \left\{ 1, \frac{\varepsilon}{1 + 2|z_0|} \right\}$;

the required is obtained.

show that

$$\lim_{z \rightarrow z_0} \operatorname{Re} z = \operatorname{Re} z_0 \quad \text{is true.}$$

because

$$|\operatorname{Re} z| \leq |z| ;$$

$$|f(z) - w_0| = |\operatorname{Re} z - \operatorname{Re} z_0| = |\operatorname{Re}(z - z_0)| \leq |z - z_0| < \varepsilon$$

$\delta = \varepsilon$ is chosen.

The required is obtained.

$$\lim_{z \rightarrow 2i} (2x + iy^2) = 4i \quad ?$$

$$0 < |z - 2i| < \delta$$

$$|2x + iy^2 - 4i| < \varepsilon$$

$$|2x + i(y^2 - 4)| < \varepsilon$$

$$|2x + i(y^2 - 4)| \leq |2x| + |y^2 - 4| = 2|x| + |y - 2||y + 2|$$

$$|2x + iy^2 - 4i| < \varepsilon \text{ is satisfied while}$$

$$2|x| < \frac{\varepsilon}{2} \text{ and}$$

$$|y - 2||y + 2| < \frac{\varepsilon}{2}$$

$$\Rightarrow |x| < \frac{\varepsilon}{4}$$

$$\Rightarrow |y + 2| = |y - 2 + 4| \leq |y - 2| + 4$$

$$\text{when } |y - 2| < 1;$$

$$|y + 2| < 5$$

$$\text{Thus, } |y - 2| < \min\left(\frac{\varepsilon}{10}, 1\right)$$

$$\text{If } \delta = \min\left(\frac{\varepsilon}{10}, 1\right), \quad 0 < |z - 2i| < \delta, \quad |f(z) - 4i| < \varepsilon \text{ is satisfied.}$$

Really,

$$|z-2i| < \delta \quad |x+iy-2i| < \delta$$

$$|x+i(y-2)| < \delta$$

$$|x| < |x+i(y-2)|$$

$$|y-2| < |x+i(y-2)|$$

$$|x| < \delta < \frac{\varepsilon}{10} < \frac{\varepsilon}{4}$$

$$|y-2| < \delta < 1 \quad |y+2| < 5$$

$$|y-2| < \delta < \frac{\varepsilon}{10} \quad |y-2| < \frac{\varepsilon}{10}$$

$$|f(z)-4i| = |2x+iy^2-4| \leq 2|x|+|y-2||y+2| < 2 \cdot \frac{\varepsilon}{4} + \frac{\varepsilon}{10} \cdot 5 = \varepsilon$$

if $f(z) = \frac{1}{z}$ $\lim_{z \rightarrow i} f(z) = -i$?

for $\varepsilon > 0$, $\exists \delta > 0$, $0 < |z - i| < \delta$, $|\frac{1}{z} + i| < \varepsilon$

$$|\frac{1}{z} + i| = |\frac{1 + iz}{z}| = \frac{|1 + iz|}{|z|} = \frac{|i||z - i|}{|z|} = \frac{|z - i|}{|z|} = \frac{\delta}{1 - \delta}$$

if $\delta = \frac{\varepsilon}{1 + \varepsilon}$ is selected, the required is obtained.

If f has limit at z_0 , it is unique.

Proof: Let us assume $\lim_{z \rightarrow z_0} f(z) = a$ $\lim_{z \rightarrow z_0} f(z) = b$

and $a \neq b$

Let $2\varepsilon = |a - b|$

$$2\varepsilon = |a - b| = |a - f(z) + f(z) + b| \leq \underbrace{|f(z) - a|}_{\varepsilon} + \underbrace{|f(z) - b|}_{\varepsilon} < 2\varepsilon$$

$2\varepsilon < 2\varepsilon$ is not possible; $a = b$

$$f(z) = \begin{cases} \frac{z^4 - z_0^4}{z - z_0} & z \neq z_0 \\ 4z^3 & z = z_0 \end{cases}$$

If the function is continuous at z_0

$$z^4 - z_0^4 = z_0^4 \left(\left(\frac{z}{z_0} \right)^4 - 1 \right) = z_0^4 (\omega^4 - 1) = z_0^4 ((\omega - 1)(\omega^3 + \omega^2 + \omega + 1))$$

$$z - z_0 = z_0 \left(\frac{z}{z_0} - 1 \right) = (\omega - 1) \cdot z_0$$

$$\omega^4 - 1 = (\omega - 1) \cdot (\omega^3 + \omega^2 + \omega + 1)$$

$$\begin{aligned} x^4 - y^4 &= (x^2 - y^2) \cdot (x^2 + y^2) \\ &= (x - y)(x + y)(x^2 + y^2) \\ &= (x - y)(x^3 + xy^2 + yx^2 + y^3) \end{aligned}$$

$$\frac{z^4 - z_0^4}{z - z_0} = z_0^3 \cdot (\omega^3 + \omega^2 + \omega + 1) \quad \text{while } z \rightarrow z_0; \omega \rightarrow 1$$

$$\lim_{z \rightarrow z_0} f(z) = \lim_{\omega \rightarrow 1} z_0^3 (\omega^3 + \omega^2 + \omega + 1) = 4z_0^3$$

it is continuous

$$f(z) = \frac{\bar{z} - \bar{z}_0}{z - z_0} \quad z_0 \neq 0 \quad \text{is given}$$

$$\begin{aligned} z - z_0 &= t e^{i\alpha} \\ \bar{z} - \bar{z}_0 &= t \cdot e^{-i\alpha} \end{aligned} \Rightarrow \frac{\bar{z} - \bar{z}_0}{z - z_0} = e^{-2i\alpha}$$

Limit of the function does not exist as

$$z \rightarrow z_0$$

Because of

α

Compute the following limits if they exist:

(a) $\lim_{z \rightarrow -i} \frac{iz^3 + 1}{z^2 + 1};$

(b) $\lim_{z \rightarrow \infty} \frac{4 + z^2}{(z - 1)^2}.$

(c) $\lim_{z \rightarrow 0} \frac{\operatorname{Im}(z)}{z}.$

Solution. (a)

$$\begin{aligned}\lim_{z \rightarrow -i} \frac{iz^3 + 1}{z^2 + 1} &= \lim_{z \rightarrow -i} \frac{i(z^3 + i^3)}{z^2 + 1} \\&= \lim_{z \rightarrow -i} \frac{i(z + i)(z^2 - iz + i^2)}{(z + i)(z - i)} \\&= \lim_{z \rightarrow -i} \frac{i(z^2 - iz + i^2)}{z - i} \\&= i \frac{\lim_{z \rightarrow -i} (z^2 - iz + i^2)}{\lim_{z \rightarrow -i} (z - i)} \\&= i \frac{\lim_{z \rightarrow -i} z^2 - i \lim_{z \rightarrow -i} z + \lim_{z \rightarrow -i} i^2}{\lim_{z \rightarrow -i} z - \lim_{z \rightarrow -i} i} \\&= \frac{i((-i)^2 - i(-i) + i^2)}{-i - i} = \frac{3}{2}\end{aligned}$$

(b)

$$\begin{aligned}\lim_{z \rightarrow \infty} \frac{4 + z^2}{(z - 1)^2} &= \lim_{z \rightarrow 0} \frac{4 + z^{-2}}{(z^{-1} - 1)^2} \\ &= \lim_{z \rightarrow 0} \frac{4z^2 + 1}{(1 - z)^2} = \frac{\lim_{z \rightarrow 0} (4z^2 + 1)}{\lim_{z \rightarrow 0} (1 - z)^2} \\ &= \frac{4(\lim_{z \rightarrow 0} z)^2 + \lim_{z \rightarrow 0} 1}{(\lim_{z \rightarrow 0} 1 - \lim_{z \rightarrow 0} z)^2} = 1\end{aligned}$$

(c) Since

$$\lim_{\operatorname{Re}(z)=0, z \rightarrow 0} \frac{\operatorname{Im}(z)}{z} = \lim_{y \rightarrow 0} \frac{y}{yi} = -i$$

and

$$\lim_{\operatorname{Im}(z)=0, z \rightarrow 0} \frac{\operatorname{Im}(z)}{z} = \lim_{x \rightarrow 0} \frac{0}{x} = 0$$

the limit does not exist.

Show the following limits:

$$(a) \lim_{z \rightarrow \infty} \frac{4z^5}{z^5 - 42z} = 4;$$

$$(b) \lim_{z \rightarrow \infty} \frac{z^4}{z^2 + 42z} = \infty;$$

$$(c) \lim_{z \rightarrow \infty} \frac{(az + b)^3}{(cz + d)^3} = \frac{a^3}{c^3}, \text{ if } c \neq 0.$$

Solution. (a)

$$\lim_{z \rightarrow \infty} \frac{4z^5}{z^5 - 42z} = \lim_{z \rightarrow 0} \frac{4 \left(\frac{1}{z}\right)^5}{\left(\frac{1}{z}\right)^5 - 42 \left(\frac{1}{z}\right)} = \lim_{z \rightarrow 0} \frac{4}{1 - 42z^4} = 4.$$

(b) Notice that

$$\begin{aligned} \lim_{z \rightarrow \infty} \frac{z^4}{z^2 + 42z} &\iff \lim_{z \rightarrow 0} \left[\frac{1/z^4}{1/z^2 + 42/z} \right]^{-1} \\ &\iff \lim_{z \rightarrow 0} \left[\frac{1}{z^2 + 42z^3} \right]^{-1} \\ &\iff \lim_{z \rightarrow 0} (z^2 + 42z^3) = 0. \end{aligned}$$

Therefore, $\lim_{z \rightarrow \infty} \frac{z^4}{z^2 + 42z} = \infty$.

(c)

$$\lim_{z \rightarrow \infty} \frac{(az + b)^3}{(cz + d)^3} = \lim_{z \rightarrow 0} \frac{(a/z + b)^3}{(c/z + d)^3} = \lim_{z \rightarrow 0} \frac{(a + bz)^3}{(c + dz)^3} = \frac{a^3}{c^3}.$$

Show that if $f(z)$ is continuous at z_0 , so is $|f(z)|$.

Proof. Let $f(z) = u(x, y) + iv(x, y)$. Since $f(z)$ is continuous at $z_0 = x_0 + y_0i$, $u(x, y)$ and $v(x, y)$ are continuous at (x_0, y_0) . Therefore,

$$(u(x, y))^2 + (v(x, y))^2$$

is continuous at (x_0, y_0) since the sums and products of continuous functions are continuous. It follows that

$$|f(z)| = \sqrt{(u(x, y))^2 + (v(x, y))^2}$$

is continuous at z_0 since the compositions of continuous functions are continuous.

$$\text{Ex. } f(z) = \begin{cases} \frac{z \cdot \operatorname{Re} z}{|z|} & , z \neq 0 \\ 0 & , z = 0 \end{cases}$$

Show that $f(z)$ is continuous at $z_0 = 0$

1) For $\forall \epsilon > 0$, there exists $\exists \delta > 0$ such that
 $|f(z) - f(z_0)| < \epsilon$ is satisfied

while $|z - z_0| < \delta$. ??

$$|f(z) - f(0)| = \left| \frac{z \cdot \operatorname{Re} z}{|z|} - 0 \right| = \frac{|z| \cdot |\operatorname{Re} z|}{|z|} = |\operatorname{Re} z|$$

Since $|\operatorname{Re} z| \leq |z|$; if δ is chosen as ϵ ; then

$$\delta = \epsilon;$$

while $|z - z_0| < \delta$; $|f(z) - f(z_0)| < \epsilon$ can be satisfied

$$2) f(z_0) = f(0) = 0$$

$$3) \lim_{z \rightarrow 0} f(z) = 0 = f(0) \text{ , so, } f(z) \text{ is continuous}$$

That is

Ex. Determine whether $f(z) = \frac{z^2 + 4z}{z^2 + 1}$ continuous

at $z_0 = i$

$g(z) = z^2 + 4z$ and $h(z) = z^2 + 1$ are continuous
in whole ^{entire} complex plane. Thus, f is continuous
except points that are satisfying $h(z) = z^2 + 1 = 0$

Ex $\lim_{z \rightarrow z_0} z^2 = z_0^2$

For $\forall \varepsilon > 0$, there exist $\exists \delta > 0$, $|z^2 - z_0^2| < \varepsilon$ is satisfied while $|z - z_0| < \delta$??

$$\begin{aligned} |z^2 - z_0^2| &= |z - z_0| |z + z_0| \\ &= |z - z_0| |z + 2z_0 - z_0| \\ &\leq |z - z_0| (|z - z_0| + |2z_0|) \end{aligned}$$

while $|z - z_0| < 1$

$$|z^2 - z_0^2| < |z - z_0| (1 + |2z_0|)$$

if we choose $\delta_1 = \frac{\varepsilon}{1 + |2z_0|}$; $\delta = \min \left\{ 1, \frac{\varepsilon}{1 + |2z_0|} \right\}$

for $\forall \varepsilon > 0$; The ^{requested} δ has been found.

EX: $R = \{z : |z| \leq 3\}$ is given. Determine whether $f(z) = z^2$ is continuous.

$$|z^2 - z_0^2| = |z - z_0| |z + z_0|$$

$$\leq |z - z_0| (|z| + |z_0|)$$

$$< \underbrace{|z - z_0|} \cdot 6$$

$\Downarrow$
if we choose $\delta = \frac{\epsilon}{6}$

$\Rightarrow |z^2 - z_0^2| < \epsilon$ is found.

Here, δ depends on only ϵ .

So, $f(z)$ is smooth
continuous function.

In real analysis we visualize a continuous function as a function whose graph has no breaks or holes in it. It is natural to ask if there is an analogous property for continuous complex functions. The answer is yes, but this property must be stated in terms of complex mappings. We begin by recalling that a parametric curve defined by parametric equations $x = x(t)$ and $y = y(t)$ is called continuous if the real functions x and y are continuous. In a similar manner, we say that a complex parametric curve defined by $z(t) = x(t) + iy(t)$ is continuous if both $x(t)$ and $y(t)$ are continuous real functions. As with parametric curves in the Cartesian plane, a continuous parametric curve in the complex plane has no breaks or holes in it. Such curves provide a means to visualize continuous complex functions.

If a complex function f is continuous on a set S , then the image of every continuous parametric curve in S must be a continuous curve.

Derivative of Complex Function

Suppose the complex function f is defined in a neighborhood of a point z_0 . The derivative of f at z_0 , denoted by $f'(z_0)$, is

$$f'(z_0) = \lim_{\Delta z \rightarrow 0} \frac{f(z_0 + \Delta z) - f(z_0)}{\Delta z} \quad (1)$$

provided this limit exists.

If the limit in (1) exists, then the function f is said to be differentiable at z_0 .

EXAMPLE 1 Using Definition

Use Definition to find the derivative of $f(z) = z^2 - 5z$.

Solution Because we are going to compute the derivative of f at any point, we replace z_0 in (1) by the symbol z . First,

$$f(z + \Delta z) = (z + \Delta z)^2 - 5(z + \Delta z) = z^2 + 2z\Delta z + (\Delta z)^2 - 5z - 5\Delta z.$$

Second,

$$\begin{aligned} f(z + \Delta z) - f(z) &= z^2 + 2z\Delta z + (\Delta z)^2 - 5z - 5\Delta z - (z^2 - 5z) \\ &= 2z\Delta z + (\Delta z)^2 - 5\Delta z. \end{aligned}$$

Then, finally, (1) gives

$$\begin{aligned} f'(z) &= \lim_{\Delta z \rightarrow 0} \frac{2z\Delta z + (\Delta z)^2 - 5\Delta z}{\Delta z} \\ &= \lim_{\Delta z \rightarrow 0} \frac{\Delta z(2z + \Delta z - 5)}{\Delta z} \\ &= \lim_{\Delta z \rightarrow 0} (2z + \Delta z - 5). \end{aligned}$$

The limit is $f'(z) = 2z - 5$.

Rules of Differentiation

The familiar rules of differentiation in the calculus of real variables carry over to the calculus of complex variables. ¹¹

Differentiation Rules

$$\text{Constant Rules: } \frac{d}{dz}c = 0 \text{ and } \frac{d}{dz}cf(z) = cf'(z) \quad (2)$$

$$\text{Sum Rule: } \frac{d}{dz}[f(z) \pm g(z)] = f'(z) \pm g'(z) \quad (3)$$

$$\text{Product Rule: } \frac{d}{dz}[f(z)g(z)] = f(z)g'(z) + f'(z)g(z) \quad (4)$$

$$\text{Quotient Rule: } \frac{d}{dz} \left[\frac{f(z)}{g(z)} \right] = \frac{g(z)f'(z) - f(z)g'(z)}{[g(z)]^2} \quad (5)$$

$$\text{Chain Rule: } \frac{d}{dz}f(g(z)) = f'(g(z))g'(z). \quad (6)$$

The **power rule** for differentiation of powers of z is also valid:

$$\frac{d}{dz}z^n = nz^{n-1}, \quad n \text{ an integer.} \quad (7)$$

Combining (7) with (6) gives the **power rule for functions**:

$$\frac{d}{dz}[g(z)]^n = n[g(z)]^{n-1}g'(z), \quad n \text{ an integer.} \quad (8)$$

EXAMPLE 2 Using the Rules of Differentiation

Differentiate:

$$(a) f(z) = 3z^4 - 5z^3 + 2z \quad (b) f(z) = \frac{z^2}{4z + 1} \quad (c) f(z) = (iz^2 + 3z)^5$$

Solution

(a) Using the power rule (7), the sum rule (3), along with (2), we obtain

$$f'(z) = 3 \cdot 4z^3 - 5 \cdot 3z^2 + 2 \cdot 1 = 12z^3 - 15z^2 + 2.$$

(b) From the quotient rule (5),

$$f'(z) = \frac{(4z + 1) \cdot 2z - z^2 \cdot 4}{(4z + 1)^2} = \frac{4z^2 + 2z}{(4z + 1)^2}.$$

(c) In the power rule for functions (8) we identify $n = 5$, $g(z) = iz^2 + 3z$, and $g'(z) = 2iz + 3$, so that

$$f'(z) = 5(iz^2 + 3z)^4(2iz + 3).$$

For a complex function f to be differentiable at a point z_0 , we know from the preceding chapter that the limit $\lim_{\Delta z \rightarrow 0} \frac{f(z_0 + \Delta z) - f(z_0)}{\Delta z}$ must exist and equal the same complex number from any direction; that is, the limit must exist regardless how Δz approaches 0. This means that in complex analysis, the requirement of differentiability of a function $f(z)$ at a point z_0 is a far greater demand than in real calculus of functions $f(x)$ where we can approach a real number x_0 on the number line from only two directions.

EXAMPLE 3 A Function That Is Nowhere Differentiable

Show that the function $f(z) = x + 4iy$ is not differentiable at any point z .

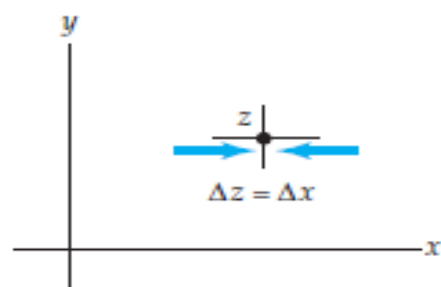
Solution Let z be any point in the complex plane. With $\Delta z = \Delta x + i\Delta y$,

$$f(z + \Delta z) - f(z) = (x + \Delta x) + 4i(y + \Delta y) - x - 4iy = \Delta x + 4i\Delta y$$

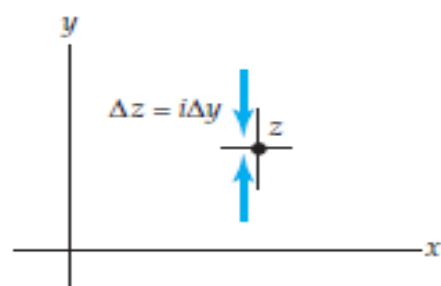
and so
$$\lim_{\Delta z \rightarrow 0} \frac{f(z + \Delta z) - f(z)}{\Delta z} = \lim_{\Delta z \rightarrow 0} \frac{\Delta x + 4i\Delta y}{\Delta x + i\Delta y}. \quad (9)$$

Now, as shown in Figure 1(a), if we let $\Delta z \rightarrow 0$ along a line parallel to the x -axis, then $\Delta y = 0$ and $\Delta z = \Delta x$ and

$$\lim_{\Delta z \rightarrow 0} \frac{f(z + \Delta z) - f(z)}{\Delta z} = \lim_{\Delta z \rightarrow 0} \frac{\Delta x}{\Delta x} = 1. \quad (10)$$



(a) $\Delta z \rightarrow 0$ along a line parallel to x -axis



(b) $\Delta z \rightarrow 0$ along a line parallel to y -axis

On the other hand, if we let $\Delta z \rightarrow 0$ along a line parallel to the y -axis as shown in Figure 1(b), then $\Delta x = 0$ and $\Delta z = i\Delta y$ so that

$$\lim_{\Delta z \rightarrow 0} \frac{f(z + \Delta z) - f(z)}{\Delta z} = \lim_{\Delta z \rightarrow 0} \frac{4i\Delta y}{i\Delta y} = 4. \quad (11)$$

In view of the obvious fact that the values in (10) and (11) are different, we conclude that $f(z) = x + 4iy$ is nowhere differentiable; that is, f is not differentiable at any point z .

Let

$$f(z) = \begin{cases} \bar{z}^3/z^2 & \text{if } z \neq 0 \\ 0 & \text{if } z = 0 \end{cases}$$

Show that

- (a) $f(z)$ is continuous everywhere on $\mathbb{C}$;
- (b) the complex derivative $f'(0)$ does not exist.

Proof. Since both $\bar{z}^3$ and z^2 are continuous on $\mathbb{C}^* = \mathbb{C} \setminus \{0\}$ and $z^2 \neq 0$, $f(z) = \bar{z}^3/z^2$ is continuous on $\mathbb{C}^*$.

At $z = 0$, we have

$$\lim_{z \rightarrow 0} |f(z)| = \lim_{z \rightarrow 0} \left| \frac{\bar{z}^3}{z^2} \right| = \lim_{z \rightarrow 0} |z| = 0$$

and hence $\lim_{z \rightarrow 0} f(z) = 0 = f(0)$. So f is also continuous at 0 and hence continuous everywhere on $\mathbb{C}$.

The complex derivative $f'(0)$, if exists, is given by

$$\lim_{z \rightarrow 0} \frac{f(z) - f(0)}{z - 0} = \lim_{z \rightarrow 0} \frac{\bar{z}^3}{z^3}.$$

Let $z = x + yi$. If $y = 0$ and $x \rightarrow 0$, then

$$\lim_{z=x \rightarrow 0} \frac{\bar{z}^3}{z^3} = \lim_{x \rightarrow 0} \frac{x^3}{x^3} = 1.$$

On the other hand, if $x = 0$ and $y \rightarrow 0$, then

$$\lim_{z=yi \rightarrow 0} \frac{\bar{z}^3}{z^3} = \lim_{y \rightarrow 0} \frac{(-yi)^3}{(yi)^3} = -1.$$

So the limit $\lim_{z \rightarrow 0} \bar{z}^3/z^3$ and hence $f'(0)$ do not exist.

Analytic Functions

A complex function $w = f(z)$ is said to be analytic at a point z_0 if f is differentiable at z_0 and at every point in some neighborhood of z_0 .

A function f is analytic in a domain D if it is analytic at every point in D . The phrase “analytic on a domain D ” is also used.

Entire Functions

A function that is analytic at every point z in the complex plane is said to be an entire function. In view of differentiation rules (2), (3), (7), and (5), we can conclude that polynomial functions are differentiable at every point z in the complex plane and rational functions are analytic throughout any domain D that contains no points at which the denominator is zero. '

Theorem Polynomial and Rational Functions

- (i) A polynomial function $p(z) = a_n z^n + a_{n-1} z^{n-1} + \cdots + a_1 z + a_0$, where n is a nonnegative integer, is an entire function.
- (ii) A rational function $f(z) = \frac{p(z)}{q(z)}$, where p and q are polynomial functions, is analytic in any domain D that contains no point z_0 for which $q(z_0) = 0$.

In general, a point z at which a complex function $w = f(z)$ fails to be analytic is called a singular point of f .

Theorem Differentiability Implies Continuity

If f is differentiable at a point z_0 in a domain D , then f is continuous at z_0 .

Proof The limits $\lim_{z \rightarrow z_0} \frac{f(z) - f(z_0)}{z - z_0}$ and $\lim_{z \rightarrow z_0} (z - z_0)$ exist and equal $f'(z_0)$ and 0, respectively.

$$\begin{aligned}\lim_{z \rightarrow z_0} (f(z) - f(z_0)) &= \lim_{z \rightarrow z_0} \frac{f(z) - f(z_0)}{z - z_0} \cdot (z - z_0) \\ &= \lim_{z \rightarrow z_0} \frac{f(z) - f(z_0)}{z - z_0} \cdot \lim_{z \rightarrow z_0} (z - z_0) = f'(z_0) \cdot 0 = 0.\end{aligned}$$

From $\lim_{z \rightarrow z_0} (f(z) - f(z_0)) = 0$ we conclude that $\lim_{z \rightarrow z_0} f(z) = f(z_0)$.

Suppose f and g are functions that are analytic at a point z_0 and $f(z_0) = 0$, $g(z_0) = 0$, but $g'(z_0) \neq 0$. Then

$$\lim_{z \rightarrow z_0} \frac{f(z)}{g(z)} = \frac{f'(z_0)}{g'(z_0)}.$$

EXAMPLE 4 Using L'Hôpital's Rule

Compute $\lim_{z \rightarrow 2+i} \frac{z^2 - 4z + 5}{z^3 - z - 10i}$.

Solution If we identify $f(z) = z^2 - 4z + 5$ and $g(z) = z^3 - z - 10i$, you should verify that $f(2+i) = 0$ and $g(2+i) = 0$. The given limit has the indeterminate form $0/0$. Now since f and g are polynomial functions, both functions are necessarily analytic at $z_0 = 2+i$. Using

$$f'(z) = 2z - 4, \quad g'(z) = 3z^2 - 1, \quad f'(2+i) = 2i, \quad g'(2+i) = 8 + 12i,$$

we see that

$$\lim_{z \rightarrow 2+i} \frac{z^2 - 4z + 5}{z^3 - z - 10i} = \frac{f'(2+i)}{g'(2+i)} = \frac{2i}{8 + 12i} = \frac{3}{26} + \frac{1}{13}i.$$

A Necessary Condition for Analyticity

In the next theorem we see that if a function $f(z) = u(x, y) + iv(x, y)$ is differentiable at a point z , then the functions u and v must satisfy a pair of equations that relate their first-order partial derivatives.

Theorem Cauchy-Riemann Equations

Suppose $f(z) = u(x, y) + iv(x, y)$ is differentiable at a point $z = x + iy$. Then at z the first-order partial derivatives of u and v exist and satisfy the **Cauchy-Riemann equations**

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \text{and} \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}. \quad (1)$$

Proof The derivative of f at z is given by

$$f'(z) = \lim_{\Delta z \rightarrow 0} \frac{f(z + \Delta z) - f(z)}{\Delta z}. \quad (2)$$

By writing $f(z) = u(x, y) + iv(x, y)$ and $\Delta z = \Delta x + i\Delta y$, (2) becomes

$$f'(z) = \lim_{\Delta z \rightarrow 0} \frac{u(x + \Delta x, y + \Delta y) + iv(x + \Delta x, y + \Delta y) - u(x, y) - iv(x, y)}{\Delta x + i\Delta y}. \quad (3)$$

Since the limit (2) is assumed to exist, Δz can approach zero from any convenient direction. In particular, if we choose to let $\Delta z \rightarrow 0$ along a horizontal line, then $\Delta y = 0$ and $\Delta z = \Delta x$. We can then write (3) as

$$\begin{aligned}
f'(z) &= \lim_{\Delta x \rightarrow 0} \frac{u(x + \Delta x, y) - u(x, y) + i[v(x + \Delta x, y) - v(x, y)]}{\Delta x} \\
&= \lim_{\Delta x \rightarrow 0} \frac{u(x + \Delta x, y) - u(x, y)}{\Delta x} + i \lim_{\Delta x \rightarrow 0} \frac{v(x + \Delta x, y) - v(x, y)}{\Delta x}.
\end{aligned} \tag{4}$$

The existence of $f'(z)$ implies that each limit in (4) exists. These limits are the definitions of the first-order partial derivatives with respect to x of u and v , respectively. Hence, we have shown two things: both $\partial u/\partial x$ and $\partial v/\partial x$ exist at the point z , and that the derivative of f is

$$f'(z) = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x}. \tag{5}$$

We now let $\Delta z \rightarrow 0$ along a vertical line. With $\Delta x = 0$ and $\Delta z = i\Delta y$, (3) becomes

$$f'(z) = \lim_{\Delta y \rightarrow 0} \frac{u(x, y + \Delta y) - u(x, y)}{i\Delta y} + i \lim_{\Delta y \rightarrow 0} \frac{v(x, y + \Delta y) - v(x, y)}{i\Delta y}. \quad (6)$$

In this case (6) shows us that $\partial u/\partial y$ and $\partial v/\partial y$ exist at z and that

$$f'(z) = -i \frac{\partial u}{\partial y} + \frac{\partial v}{\partial y}. \quad (7)$$

By equating the real and imaginary parts of (5) and (7) we obtain the pair of equations in (1).

Because Theorem 1 states that the Cauchy-Riemann equations (1) hold at z as a *necessary* consequence of f being differentiable at z , we cannot use the theorem to help us determine where f is differentiable. *But* it is important to realize that Theorem 1 can tell us where a function f does not possess a derivative. If the equations in (1) are *not* satisfied at a point z , then f cannot be differentiable at z .

$f(z) = x + 4iy$ is not differentiable at any point z . If we identify $u = x$ and $v = 4y$, then $\partial u/\partial x = 1$, $\partial v/\partial y = 4$, $\partial u/\partial y = 0$, and $\partial v/\partial x = 0$. In view of

$$\frac{\partial u}{\partial x} = 1 \neq \frac{\partial v}{\partial y} = 4$$

the two equations in (1) cannot be simultaneously satisfied at any point z . In other words, f is nowhere differentiable.

if a complex function $f(z) = u(x, y) + iv(x, y)$ is analytic throughout a domain D , then the real functions u and v satisfy the Cauchy-Riemann equations (1) at every point in D .

EXAMPLE 1

The polynomial function $f(z) = z^2 + z$ is analytic for all z and can be written as $f(z) = x^2 - y^2 + x + i(2xy + y)$. Thus, $u(x, y) = x^2 - y^2 + x$ and $v(x, y) = 2xy + y$. For any point (x, y) in the complex plane we see that the Cauchy-Riemann equations are satisfied:

$$\frac{\partial u}{\partial x} = 2x + 1 = \frac{\partial v}{\partial y} \quad \text{and} \quad \frac{\partial u}{\partial y} = -2y = -\frac{\partial v}{\partial x}.$$

Criterion for Non-analyticity

If the Cauchy-Riemann equations are not satisfied at every point z in a domain D , then the function $f(z) = u(x, y) + iv(x, y)$ cannot be analytic in D .

EXAMPLE 2 Using the Cauchy-Riemann Equations

Show that the complex function $f(z) = 2x^2 + y + i(y^2 - x)$ is not analytic at any point.

Solution We identify $u(x, y) = 2x^2 + y$ and $v(x, y) = y^2 - x$. From

$$\begin{aligned}\frac{\partial u}{\partial x} &= 4x & \text{and} & & \frac{\partial v}{\partial y} &= 2y \\ \frac{\partial u}{\partial y} &= 1 & \text{and} & & \frac{\partial v}{\partial x} &= -1\end{aligned}\tag{8}$$

we see that $\partial u/\partial y = -\partial v/\partial x$ but that the equality $\partial u/\partial x = \partial v/\partial y$ is satisfied only on the line $y = 2x$. However, for any point z on the line, there is no neighborhood or open disk about z in which f is differentiable at every point. We conclude that f is nowhere analytic.

A Sufficient Condition for Analyticity

By themselves, the Cauchy-Riemann equations do not ensure analyticity of a function $f(z) = u(x, y) + iv(x, y)$ at a point $z = x + iy$. It is possible for the Cauchy-Riemann equations to be satisfied at z and yet $f(z)$ may not be differentiable at z , or $f(z)$ may be differentiable at z but nowhere else. In either case, f is not analytic at z .

However, when we add the condition of continuity to u and v and to the four partial derivatives $\partial u/\partial x$, $\partial u/\partial y$, $\partial v/\partial x$, and $\partial v/\partial y$, it can be shown that the Cauchy-Riemann equations are not only necessary but also sufficient to guarantee analyticity of $f(z) = u(x, y) + iv(x, y)$ at z . The proof is long and complicated and so we state only the result.

Theorem Criterion for Analyticity

Suppose the real functions $u(x, y)$ and $v(x, y)$ are continuous and have continuous first-order partial derivatives in a domain D . If u and v satisfy the Cauchy-Riemann equations (1) at all points of D , then the complex function $f(z) = u(x, y) + iv(x, y)$ is analytic in D .

EXAMPLE 3 Using Theorem

For the function $f(z) = \frac{x}{x^2 + y^2} - i\frac{y}{x^2 + y^2}$, the real functions $u(x, y) = \frac{x}{x^2 + y^2}$ and $v(x, y) = -\frac{y}{x^2 + y^2}$ are continuous except at the point where $x^2 + y^2 = 0$, that is, at $z = 0$. Moreover, the first four first-order partial derivatives

$$\begin{aligned}\frac{\partial u}{\partial x} &= \frac{y^2 - x^2}{(x^2 + y^2)^2}, & \frac{\partial u}{\partial y} &= -\frac{2xy}{(x^2 + y^2)^2}, \\ \frac{\partial v}{\partial x} &= \frac{2xy}{(x^2 + y^2)^2}, & \text{and} & \frac{\partial v}{\partial y} = \frac{y^2 - x^2}{(x^2 + y^2)^2}\end{aligned}$$

are continuous except at $z = 0$. Finally, we see from

$$\frac{\partial u}{\partial x} = \frac{y^2 - x^2}{(x^2 + y^2)^2} = \frac{\partial v}{\partial y} \quad \text{and} \quad \frac{\partial u}{\partial y} = -\frac{2xy}{(x^2 + y^2)^2} = -\frac{\partial v}{\partial x}$$

that the Cauchy-Riemann equations are satisfied except at $z = 0$. Thus we conclude from Theorem that f is analytic in any domain D that does not contain the point $z = 0$.

In other words, (5) and (7) give us a formula for computing the derivative $f'(z)$:

$$f'(z) = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} = \frac{\partial v}{\partial y} - i \frac{\partial u}{\partial y}. \quad (9)$$

Sufficient Conditions for Differentiability

If the real functions $u(x,y)$ and $v(x, y)$ are continuous and have continuous first-order partial derivatives in some neighborhood of a point z , and if u and v satisfy the Cauchy-Riemann equations (1) at z , then the complex function $f(z) = u(x, y) + iv(x, y)$ is differentiable at z and $f'(z)$ is given by (9).

EXAMPLE 4 A Function Differentiable on a Line

the complex function $f(z) = 2x^2 + y + i(y^2 - x)$ was nowhere analytic, but yet the Cauchy-Riemann equations were satisfied on the line $y = 2x$. But since the functions $u(x, y) = 2x^2 + y$, $\partial u/\partial x = 4x$, $\partial u/\partial y = 1$, $v(x, y) = y^2 - x$, $\partial v/\partial x = -1$ and $\partial v/\partial y = 2y$ are continuous at every point, it follows that f is differentiable on the line $y = 2x$. Moreover, from (9) we see that the derivative of f at points on this line is given by $f'(z) = 4x - i = 2y - i$.

Suppose the function $f(z) = u(x, y) + iv(x, y)$ is analytic in a domain D .

- (i) If $|f(z)|$ is constant in D , then so is $f(z)$.
- (ii) If $f'(z) = 0$ in D , then $f(z) = c$ in D , where c is a constant.

The Cauchy–Riemann equations also help in deriving general properties of analytic functions.

For instance, show that if $f(z)$ is analytic in a domain D and $|f(z)| = k = \text{const}$ in D , then $f(z) = \text{const}$ in D .

Solution. By assumption, $|f|^2 = |u + iv|^2 = u^2 + v^2 = k^2$. By differentiation,

$$uu_x + vv_x = 0,$$

$$uu_y + vv_y = 0.$$

Now use $v_x = -u_y$ in the first equation and $v_y = u_x$ in the second, to get

$$(a) \quad uu_x - vu_y = 0,$$

$$(b) \quad uu_y + vu_x = 0.$$

To get rid of u_y , multiply (a) by u and (b) by v and add. Similarly, to eliminate u_x , multiply (a) by $-v$ and by u and add. This yields

$$(u^2 + v^2)u_x = 0,$$

$$(u^2 + v^2)u_y = 0.$$

If $k^2 = u^2 + v^2 = 0$, then $u = v = 0$; hence $f = 0$. If $k^2 = u^2 + v^2 \neq 0$, then $u_x = u_y = 0$. Hence, by the Cauchy–Riemann equations, also $v_x = v_y = 0$. Together this implies $u = \text{const}$ and $v = \text{const}$; hence $f = \text{const}$.

Polar Coordinates

As we saw that a complex function can be expressed in terms of polar coordinates. Indeed, the form $f(z) = u(r, \theta) + iv(r, \theta)$ is often more convenient to use. In polar coordinates the Cauchy-Riemann equations become

$$\frac{\partial u}{\partial r} = \frac{1}{r} \frac{\partial v}{\partial \theta}, \quad \frac{\partial v}{\partial r} = -\frac{1}{r} \frac{\partial u}{\partial \theta}. \quad (10)$$

The polar version of (9) at a point z whose polar coordinates are (r, θ) is then

$$f'(z) = e^{-i\theta} \left(\frac{\partial u}{\partial r} + i \frac{\partial v}{\partial r} \right) = \frac{1}{r} e^{-i\theta} \left(\frac{\partial v}{\partial \theta} - i \frac{\partial u}{\partial \theta} \right). \quad (11)$$

$$\text{Ex: } f(z) = \frac{1}{z} = \frac{1}{re^{i\theta}} = \frac{1}{r} \cdot e^{-i\theta} = \frac{1}{r}(\cos\theta - i\sin\theta)$$

$$z \neq 0.$$

$$u(r, \theta) = \frac{\cos\theta}{r} \quad v(r, \theta) = -\frac{\sin\theta}{r}$$

$$r \cdot u_r = -\frac{\cos\theta}{r} = v_\theta \quad u_\theta = -\frac{\sin\theta}{r} = r v_r$$

If $z \neq 0$ it has derivative

$$f'(z) = e^{-i\theta} \left[-\frac{\cos\theta}{r^2} + i \frac{\sin\theta}{r^2} \right] = e^{-i\theta} \cdot \frac{e^{-i\theta}}{r^2}$$

$$= -\frac{1}{(re^{i\theta})^2} = -\frac{1}{z^2}$$

$$\boxed{r \cdot u_r = v_\theta}$$

$$\boxed{u_\theta = -r v_r}$$

$$\forall z \in \mathbb{C}: f(z) = \sqrt[3]{r} e^{i\theta/3} \quad r > 0 \quad \alpha < \theta < \alpha + 2\pi \quad \alpha \in \mathbb{R}$$

$$u(r, \theta) = \sqrt[3]{r} \cos \frac{\theta}{3} \quad v(r, \theta) = \sqrt[3]{r} \sin \frac{\theta}{3}$$

$$\bullet \quad r u_r = \frac{\sqrt[3]{r}}{3} \cos \frac{\theta}{3} = v_\theta \quad u_\theta = -\frac{\sqrt[3]{r}}{3} \sin \frac{\theta}{3} = -r v_r$$

$$f'(z) = e^{-i\theta} \left[\frac{1}{3(\sqrt[3]{r})^2} \cos \frac{\theta}{3} + i \frac{1}{3(\sqrt[3]{r})^2} \sin \frac{\theta}{3} \right]$$

$$f'(z) = \frac{e^{-i\theta}}{3(\sqrt[3]{r})^2} e^{i\theta/3} = \frac{1}{3(\sqrt[3]{r} \cdot e^{i\theta/3})^2} = \frac{1}{3(f(z))^2}$$

The $\partial f / \partial \bar{z} = 0$ form of the Cauchy Riemann equations

Recall that we showed that $f(z) = \bar{z}$ is not analytic anywhere and more generally if $f(z)$ is analytic then $g(z) = f(\bar{z})$ is not complex differentiable at any point for which $f'(\bar{z}) \neq 0$. These observations all suggest that for f to be analytic then it should be possible to express it in a way which does not involve $\bar{z}$. This is indeed the case in the following sense.

From $\underline{z = x + iy}$ and $\underline{\bar{z} = x - iy}$ we have

$$x = \frac{z + \bar{z}}{2}, \quad y = \frac{z - \bar{z}}{2i}$$

and we can write

$$g(z, \bar{z}) = u(x, y) + iv(x, y) = u\left(\frac{z + \bar{z}}{2}, \frac{z - \bar{z}}{2i}\right) + iv\left(\frac{z + \bar{z}}{2}, \frac{z - \bar{z}}{2i}\right).$$

For the manipulations we can consider g as independent function of z and $\bar{z}$ and use the chain rule to give

$$\begin{aligned}\frac{\partial g}{\partial \bar{z}} &= \left(\frac{\partial u}{\partial x} \frac{\partial x}{\partial \bar{z}} + \frac{\partial u}{\partial y} \frac{\partial y}{\partial \bar{z}} \right) + i \left(\frac{\partial v}{\partial x} \frac{\partial x}{\partial \bar{z}} + \frac{\partial v}{\partial y} \frac{\partial y}{\partial \bar{z}} \right) \\ &= \left(\frac{1}{2} \frac{\partial u}{\partial x} - \frac{1}{2i} \frac{\partial u}{\partial y} \right) + i \left(\frac{1}{2} \frac{\partial v}{\partial x} - \frac{1}{2i} \frac{\partial v}{\partial y} \right) \\ &= \left(\frac{\partial u}{\partial x} - \frac{\partial v}{\partial y} \right) - \frac{1}{2i} \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right).\end{aligned}$$

This shows that

$$\frac{\partial g}{\partial \bar{z}} = 0 \quad \text{if and only if} \quad \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \text{and} \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}.$$

It is in this sense that if we have a function of z and $\bar{z}$ then it needs to be a function of z alone to be analytic. This does not mean that we can always express an analytic function as a finite combination of standard functions as in all the examples considered so far but as we will show later in the module we can always represent such functions in a power series about a point z_0 which just involves powers of $z - z_0$.

Harmonic Functions

we shall see that when a complex function $f(z) = u(x, y) + iv(x, y)$ is analytic at a point z , then *all* the derivatives of f : $f'(z), f''(z), f'''(z), \dots$ are also analytic at z . As a consequence of this remarkable fact, we can conclude that *all* partial derivatives of the real functions $u(x, y)$ and $v(x, y)$ are continuous at z . From the continuity of the partial derivatives we then know that the second-order mixed partial derivatives are equal. This last fact, coupled with the Cauchy-Riemann equations, will be used in this section to demonstrate that there is a connection between the real and imaginary parts of an analytic function $f(z) = u(x, y) + iv(x, y)$ and the second-order partial differential equation

$$\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = 0. \quad (1)$$

This equation, one of the most famous in applied mathematics, is known as **Laplace's equation** in two variables. The sum $\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2}$ of the two second partial derivatives in (1) is denoted by $\nabla^2 \phi$ and is called the **Laplacian** of ϕ . Laplace's equation is then abbreviated as $\nabla^2 \phi = 0$.

Harmonic Functions A solution $\phi(x, y)$ of Laplace's equation (1) in a domain D of the plane is given a special name.

Definition

A real-valued function ϕ of two real variables x and y that has continuous first and second-order partial derivatives in a domain D and satisfies Laplace's equation is said to be **harmonic** in D .

Theorem **Harmonic Functions**

Suppose the complex function $f(z) = u(x, y) + iv(x, y)$ is analytic in a domain D . Then the functions $u(x, y)$ and $v(x, y)$ are harmonic in D .

Proof Assume $f(z) = u(x, y) + iv(x, y)$ is analytic in a domain D and that u and v have continuous second-order partial derivatives in D .[†] Since f is analytic, the Cauchy-Riemann equations are satisfied at every point z . Differentiating both sides of $\partial u/\partial x = \partial v/\partial y$ with respect to x and differentiating both sides of $\partial u/\partial y = -\partial v/\partial x$ with respect to y give, respectively,

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 v}{\partial x \partial y} \quad \text{and} \quad \frac{\partial^2 u}{\partial y^2} = -\frac{\partial^2 v}{\partial y \partial x}. \quad (2)$$

With the assumption of continuity, the mixed partials $\partial^2 v/\partial x \partial y$ and $\partial^2 v/\partial y \partial x$ are equal. Hence, by adding the two equations in (2) we see that

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 \quad \text{or} \quad \nabla^2 u = 0.$$

This shows that $u(x, y)$ is harmonic.

Now differentiating both sides of $\partial u/\partial x = \partial v/\partial y$ with respect to y and differentiating both sides of $\partial u/\partial y = -\partial v/\partial x$ with respect to x , give, in turn, $\partial^2 u/\partial y \partial x = \partial^2 v/\partial y^2$ and $\partial^2 u/\partial x \partial y = -\partial^2 v/\partial^2 x$. Subtracting the last two equations yields $\nabla^2 v = 0$.

EXAMPLE 1 Harmonic Functions

The function $f(z) = z^2 = x^2 - y^2 + 2xyi$ is entire. The functions $u(x, y) = x^2 - y^2$ and $v(x, y) = 2xy$ are necessarily harmonic in any domain D of the complex plane.

Harmonic Conjugate Functions

We have just shown that if a function $f(z) = u(x, y) + iv(x, y)$ is analytic in a domain D , then its real and imaginary parts u and v are necessarily harmonic in D . Now suppose $u(x, y)$ is a given real function that is known to be harmonic in D . If it is possible to find another real harmonic function $v(x, y)$ so that u and v satisfy the Cauchy-Riemann equations throughout the domain D , then the function $v(x, y)$ is called a **harmonic conjugate** of $u(x, y)$. By combining the functions as $u(x, y) + iv(x, y)$ we obtain a function that is analytic in D .

EXAMPLE 2 Harmonic Conjugate

- (a) Verify that the function $u(x, y) = x^3 - 3xy^2 - 5y$ is harmonic in the entire complex plane.
- (b) Find the harmonic conjugate function of u .

Solution

- (a) From the partial derivatives

$$\frac{\partial u}{\partial x} = 3x^2 - 3y^2, \quad \frac{\partial^2 u}{\partial x^2} = 6x, \quad \frac{\partial u}{\partial y} = -6xy - 5, \quad \frac{\partial^2 u}{\partial y^2} = -6x$$

we see that u satisfies Laplace's equation

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 6x - 6x = 0.$$

- (b) Since the conjugate harmonic function v must satisfy the Cauchy-Riemann equations $\partial v / \partial y = \partial u / \partial x$ and $\partial v / \partial x = -\partial u / \partial y$, we must have

$$\frac{\partial v}{\partial y} = 3x^2 - 3y^2 \quad \text{and} \quad \frac{\partial v}{\partial x} = 6xy + 5. \quad (3)$$

Partial integration of the first equation in (3) with respect to the variable y gives $v(x, y) = 3x^2y - y^3 + h(x)$. The partial derivative with respect to x of this last equation is

$$\frac{\partial v}{\partial x} = 6xy + h'(x).$$

When this result is substituted into the second equation in (3) we obtain $h'(x) = 5$, and so $h(x) = 5x + C$, where C is a real constant. Therefore, the harmonic conjugate of u is $v(x, y) = 3x^2y - y^3 + 5x + C$.

In Example 2, by combining u and its harmonic conjugate v as $u(x, y) + iv(x, y)$, the resulting complex function

$$f(z) = x^3 - 3xy^2 - 5y + i(3x^2y - y^3 + 5x + C)$$

is an analytic function throughout the domain D consisting, in this case, of the entire complex plane.

Complex Exponential Function

Definition Complex Exponential Function

The function e^z defined by

$$e^z = e^x \cos y + ie^x \sin y \quad (1)$$

is called the complex exponential function.

$$e^{x+0i} = e^x (\cos 0 + i \sin 0) = e^x (1 + i \cdot 0) = e^x. \quad (2)$$

Theorem Analyticity of e^z

The exponential function e^z is entire and its derivative is given by:

$$\frac{d}{dz} e^z = e^z. \quad (3)$$

EXAMPLE Derivatives of Exponential Functions

Find the derivative of each of the following functions:

(a) $iz^4 (z^2 - e^z)$ and (b) $e^{z^2-(1+i)z+3}$.

Solution (a) Using (3) and the product rule

$$\begin{aligned}\frac{d}{dz} [iz^4 (z^2 - e^z)] &= iz^4 (2z - e^z) + 4iz^3 (z^2 - e^z) \\ &= 6iz^5 - iz^4 e^z - 4iz^3 e^z.\end{aligned}$$

(b) Using (3) and the chain rule

$$\frac{d}{dz} [e^{z^2-(1+i)z+3}] = e^{z^2-(1+i)z+3} \cdot (2z - 1 - i).$$

Modulus, Argument, and Conjugate The modulus, argument, and conjugate of the exponential function are easily determined from (1). If we express the complex number $w = e^z$ in polar form:

$$w = e^x \cos y + ie^x \sin y = r (\cos \theta + i \sin \theta),$$

then we see that $r = e^x$ and $\theta = y + 2n\pi$, for $n = 0, \pm 1, \pm 2, \dots$. Because r is the modulus and θ is an argument of w , we have:

$$|e^z| = e^x \tag{4}$$

and $\arg(e^z) = y + 2n\pi, n = 0, \pm 1, \pm 2, \dots$. (5)

We know from calculus that $e^x > 0$ for all real x , and so it follows from (4) that $|e^z| > 0$. This implies that $e^z \neq 0$ for all complex z . Put another way, the point $w = 0$ is not in the range of the complex function $w = e^z$. Equation (4) does not, however, rule out the possibility that e^z is a negative real number. In fact, you should verify that if, say, $z = \pi i$, then $e^{\pi i}$ is real and $e^{\pi i} < 0$.

A formula for the conjugate of the complex exponential e^z is found using properties of the real cosine and sine functions. Since the real cosine function is even, we have $\cos y = \cos(-y)$ for all y , and since the real sine function is odd, we have $-\sin y = \sin(-y)$ for all y , and so:

$$\overline{e^z} = e^x \cos y - ie^x \sin y = e^x \cos(-y) + ie^x \sin(-y) = e^{x-iy} = e^{\bar{z}}.$$

Therefore, for all complex z , we have shown:

$$\overline{e^z} = e^{\bar{z}}. \tag{6}$$

Theorem Algebraic Properties of e^z

If z_1 and z_2 are complex numbers, then

$$(i) \quad e^0 = 1$$

$$(ii) \quad e^{z_1} e^{z_2} = e^{z_1 + z_2}$$

$$(iii) \quad \frac{e^{z_1}}{e^{z_2}} = e^{z_1 - z_2}$$

$$(iv) \quad (e^{z_1})^n = e^{nz_1}, \quad n = 0, \pm 1, \pm 2, \dots$$

Periodicity

The most striking difference between the real and complex exponential functions is the periodicity of e^z . Analogous to real periodic functions, we say that a complex function f is **periodic** with period T if $f(z + T) = f(z)$ for all complex z . The real exponential function is not periodic, but the complex exponential function is because it is defined using the real cosine and sine functions, which are periodic. In particular,

we have $e^{z+2\pi i} = e^z e^{2\pi i} = e^z (\cos 2\pi + i \sin 2\pi)$. Since $\cos 2\pi = 1$ and $\sin 2\pi = 0$, this simplifies to:

$$e^{z+2\pi i} = e^z. \tag{8}$$

The complex exponential function e^z is periodic with a pure imaginary period $2\pi i$.

That is, for the function $f(z) = e^z$, we have $f(z + 2\pi i) = f(z)$ for all z .

Because (8) holds for all values of z we also have $e^{(z+2\pi i)+2\pi i} = e^{z+2\pi i}$. This fact combined with (8) implies that $e^{z+4\pi i} = e^z$. Now by repeating this process we find that $e^{z+2n\pi i} = e^z$ for $n = 0, \pm 1, \pm 2, \dots$. Thus, $-2\pi i, 4\pi i, 6\pi i$, and so on, are also periods of e^z . Furthermore, if the complex exponential function e^z maps the point z onto the point w , then it also maps the points $z \pm 2\pi i, z \pm 4\pi i, z \pm 6\pi i$, and so on, onto the point w . Thus, the complex exponential function is not one-to-one, and all values e^z are assumed in any infinite horizontal strip of width 2π in the z -plane. That is, all values of the function e^z are assumed in the set $-\infty < x < \infty, y_0 < y \leq y_0 + 2\pi$, where y_0 is a real constant. In Figure 1 we divide the complex plane into horizontal strips obtained by setting y_0 equal to any odd multiple of π . If the point z is in the infinite horizontal strip $-\infty < x < \infty, -\pi < y \leq \pi$, shown in color in

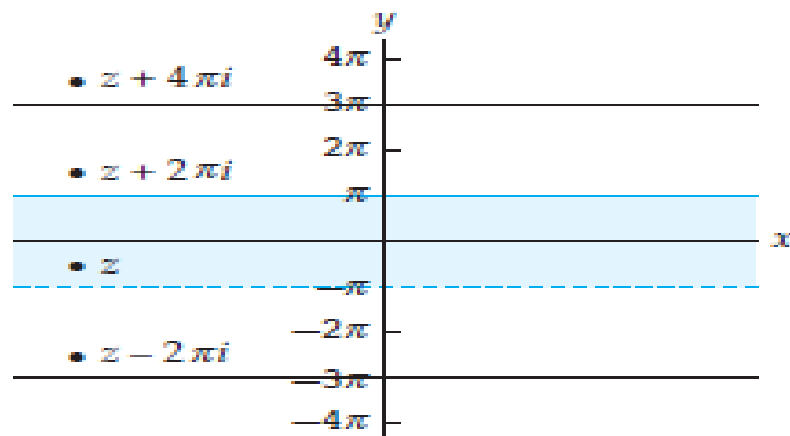


Figure 1, then the values $f(z) = e^z$, $f(z+2\pi i) = e^{z+2\pi i}$, $f(z-2\pi i) = e^{z-2\pi i}$, and so on are the same. The infinite horizontal strip defined by:

$$-\infty < x < \infty, \quad -\pi < y \leq \pi, \quad (9)$$

is called the fundamental region of the complex exponential function.

The Exponential Mapping

Because all values of the complex exponential function e^z are assumed in the fundamental region defined by (9), the image of this region under the mapping $w = e^z$ is the same as the image of the entire complex plane. In order to determine the image of the fundamental region under $w = e^z$, we note that this region consists of the collection of vertical line segments $z(t) = a + it$, $-\pi < t \leq \pi$, where a is any real number.

the image of the line segment $z(t) = a + it$, $-\pi < t \leq \pi$, under the exponential mapping is parametrized by $w(t) = e^{z(t)} = e^{a+it} = e^a e^{it}$, $-\pi < t \leq \pi$,

we see that $w(t)$ defines a circle centered at the origin with radius e^a . Because a can be any real number, the radius e^a of this circle can be any nonzero positive real number. Thus, the image of the fundamental region under the exponential mapping consists of the collection of all circles centered at the origin with nonzero radius. In other words, the image of the fundamental region $-\infty < x < \infty$, $-\pi < y \leq \pi$, under $w = e^z$ is the set of all complex w with $w \neq 0$, or, equivalently, the set $|w| > 0$.*

following

The mapping $w = e^z$ of the fundamental region is shown in Figure 1. Each vertical line segment shown in color in Figure 1(a) is mapped onto a circle shown in black in Figure 1(b) by $w = e^z$. As the x -intercept of a vertical line segment increases, the radius of its image increases. Therefore, the leftmost line segment maps onto the innermost circle, the middle line segment maps onto the middle circle, and the rightmost line segment maps onto the outermost circle.

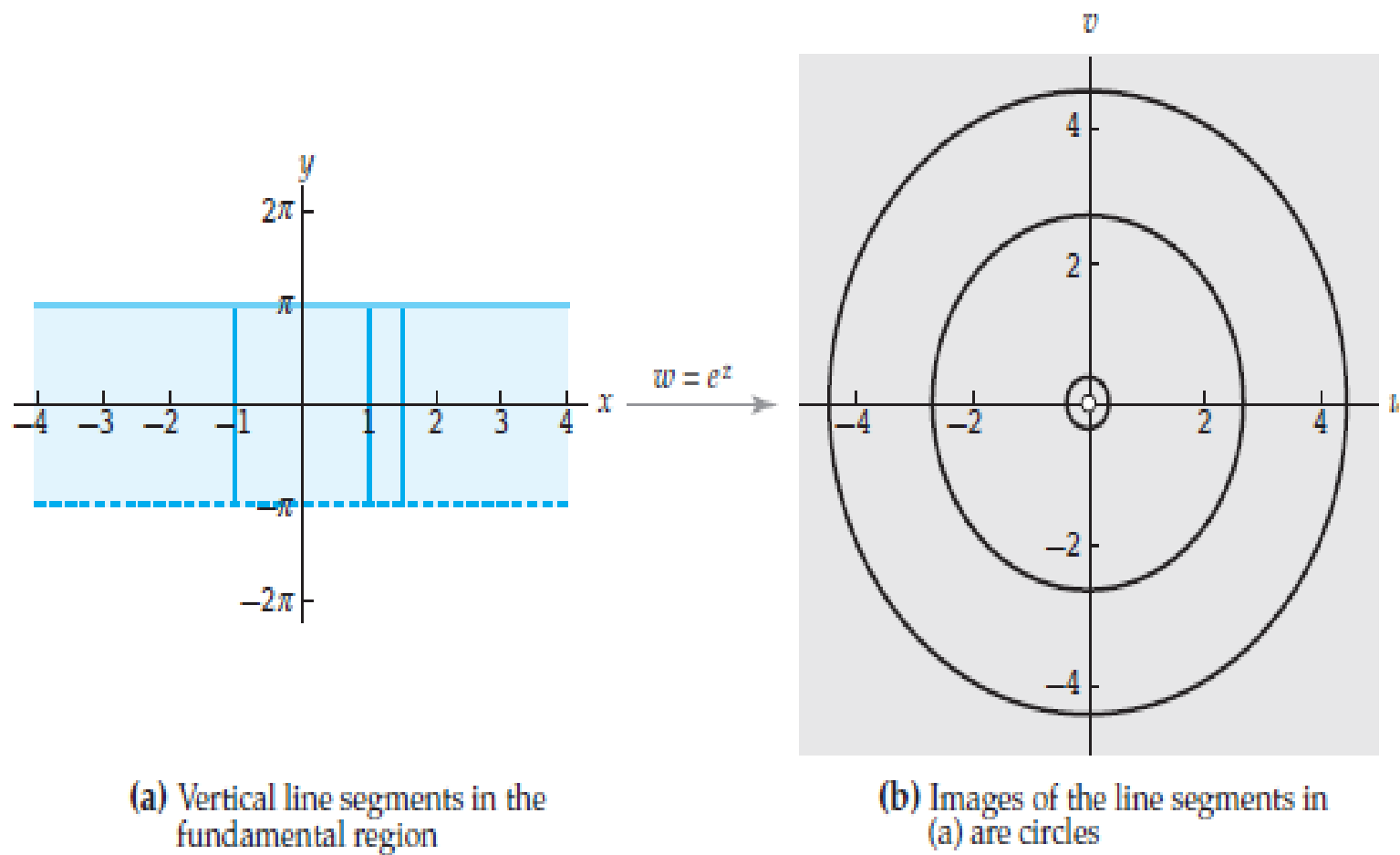
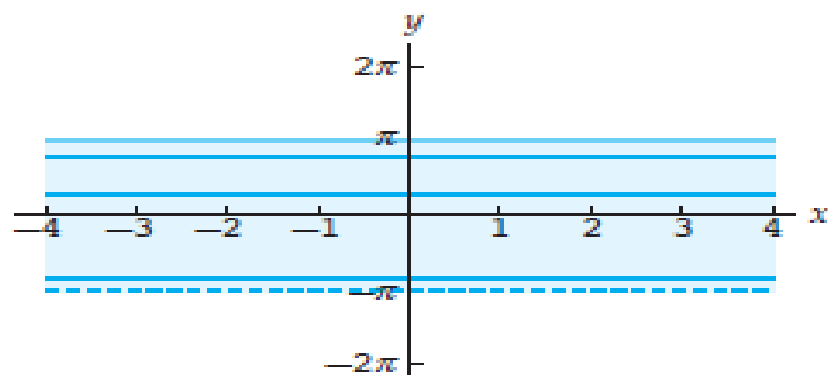


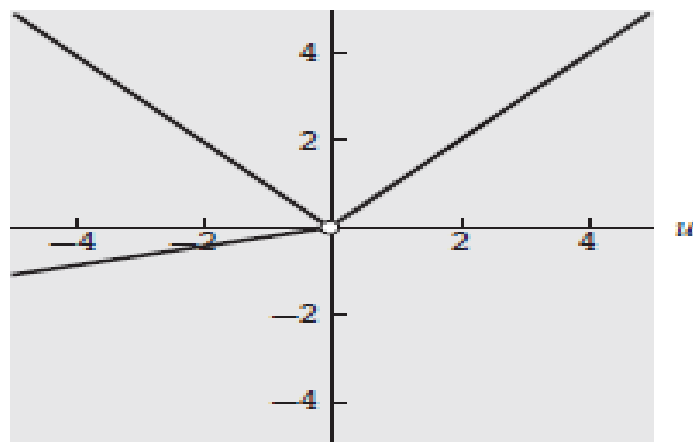
Figure The image of the fundamental region under $w = e^z$

There was nothing particularly special about using vertical line segments to determine the image of the fundamental region under $w = e^z$. The image can also be found in the same manner by using, say, horizontal lines in the fundamental region. In order to see that this is so, consider the horizontal line $y = b$. This line can be parametrized by $z(t) = t + ib$, $-\infty < t < \infty$, and so its image under $w = e^z$ is given by $w(t) = e^{z(t)} = e^{t+ib} = e^t e^{ib}$, $-\infty < t < \infty$. Define a new parameter by $s = e^t$, and observe that $0 < s < \infty$ since $-\infty < t < \infty$. Using the parameter s , the image is given by $W(s) = e^{ib}s$, $0 < s < \infty$, is the set consisting of all points $w \neq 0$ in the ray emanating from the origin and containing the point $e^{ib} = \cos b + i \sin b$. Thus, we have shown that the image of the horizontal line $y = b$ under the mapping $w = e^z$ is the set of all points $w \neq 0$ in the ray emanating from the origin and making an angle of b radians with the positive u -axis. The image can also be described by the equation $\arg(w) = b$. We represent this property of the complex exponential mapping in Figure . Each of the horizontal lines shown in color in Figure (a) is mapped onto a ray shown in black in Figure (b). As the y -intercept of a horizontal line increases, the angle the image ray makes with the positive u -axis increases. Therefore, the bottom-most line maps onto the ray in the third quadrant, the middle line maps onto the ray in the first quadrant, and the top-most line maps onto the ray in the second quadrant.



(a) Horizontal lines in the fundamental region

$w = e^z$
↓



(b) Images of the lines in (a)

Figure The mapping $w = e^z$

Exponential Mapping Properties

- (i) $w = e^z$ maps the fundamental region $-\infty < x < \infty$, $-\pi < y \leq \pi$, onto the set $|w| > 0$.
- (ii) $w = e^z$ maps the vertical line segment $x = a$, $-\pi < y \leq \pi$, onto the circle $|w| = e^a$.
- (iii) $w = e^z$ maps the horizontal line $y = b$, $-\infty < x < \infty$, onto the ray $\arg(w) = b$.

By using $\omega = f(z) = e^z$, find the mapping of the region $0 \leq x \leq \log_e 2$, $-\frac{\pi}{4} \leq y \leq \frac{\pi}{4}$ and sketch it.

(a) $0 \leq x \leq \log_e 2$ $-\frac{\pi}{4} \leq y \leq \frac{\pi}{4}$

$$\omega = e^z = e^{x+iy} = e^x (\cos y + i \sin y) \Rightarrow |\omega| = e^x$$

$$\arg(\omega) = y$$

$$\Rightarrow x=0 \Rightarrow \omega = e^0 = 1$$

$$y = -\frac{\pi}{4} \Rightarrow \arg(\omega) = -\frac{\pi}{4}$$

$$x = \log_e 2 \Rightarrow \omega = e^{\log_e 2} = 2$$

$$y = \frac{\pi}{4} \Rightarrow \arg(\omega) = \frac{\pi}{4}$$



$$\text{If } e^w = z, \text{ then } w = \log_e |z| + i \arg(z). \quad (10)$$

Because there are infinitely many arguments of z , (10) gives infinitely many solutions w to the equation $e^w = z$.¹

Definition **Complex Logarithm**

The multiple-valued function $\ln z$ defined by:

$$\ln z = \log_e |z| + i \arg(z) \quad (11)$$

is called the **complex logarithm**.

we obtain the following alternative description of the complex logarithm:

$$\ln z = \log_e r + i(\theta + 2n\pi), \quad n = 0, \pm 1, \pm 2, \dots \quad (12)$$

EXAMPLE Solving Exponential Equations

Find all complex solutions to each of the following equations.

(a) $e^w = i$ (b) $e^w = 1 + i$ (c) $e^w = -2$

Solution For each equation $e^w = z$, the set of solutions is given by $w = \ln z$ where $\ln z$ is found using Definition

(a) For $z = i$, we have $|z| = 1$ and $\arg(z) = \pi/2 + 2n\pi$. Thus, from (11) we obtain:

$$w = \ln i = \log_e 1 + i \left(\frac{\pi}{2} + 2n\pi \right).$$

Since $\log_e 1 = 0$, this simplifies to:

$$w = \frac{(4n + 1)\pi}{2}i, \quad n = 0, \pm 1, \pm 2 \dots$$

Therefore, each of the values:

$$w = \dots, -\frac{7\pi}{2}i, -\frac{3\pi}{2}i, \frac{\pi}{2}i, \frac{5\pi}{2}i, \dots$$

satisfies the equation $e^w = i$.

(b) For $z = 1 + i$, we have $|z| = \sqrt{2}$ and $\arg(z) = \pi/4 + 2n\pi$. Thus, from (11) we obtain:

$$w = \ln(1 + i) = \log_e \sqrt{2} + i \left(\frac{\pi}{4} + 2n\pi \right).$$

Because $\log_e \sqrt{2} = \frac{1}{2} \log_e 2$, this can be rewritten as:

$$w = \frac{1}{2} \log_e 2 + \frac{(8n + 1)\pi}{4} i, \quad n = 0, \pm 1, \pm 2, \dots$$

Each value of w is a solution to $e^w = 1 + i$.

(c) Again we use (11). Since $z = -2$, we have $|z| = 2$ and $\arg(z) = \pi + 2n\pi$, and so:

$$w = \ln(-2) = \log_e 2 + i(\pi + 2n\pi).$$

That is,

$$w = \log_e 2 + (2n + 1)\pi i, \quad n = 0, \pm 1, \pm 2, \dots$$

Each value of w satisfies the equation $e^w = -2$.

Theorem Algebraic Properties of $\ln z$

If z_1 and z_2 are nonzero complex numbers and n is an integer, then

$$(i) \quad \ln(z_1 z_2) = \ln z_1 + \ln z_2$$

$$(ii) \quad \ln\left(\frac{z_1}{z_2}\right) = \ln z_1 - \ln z_2$$

$$(iii) \quad \ln z_1^n = n \ln z_1.$$

Principal Value of a Complex Logarithm

It is interesting to note that the *complex* logarithm of a positive real number has infinitely many values. For example, the *complex* logarithm $\ln 5$ is the set of values $1.6094 + 2n\pi i$, where n is any integer, whereas the *real* logarithm $\log_e 5$ has a single value: $\log_e 5 \approx 1.6094$. The unique value of $\ln 5$ corresponding to $n = 0$ is the same as the value of the real logarithm $\log_e 5$. In general, this value of the complex logarithm is called the **principal value of the complex logarithm** since it is found by using the principal argument $\text{Arg}(z)$ in place of the argument $\arg(z)$ in (11). We denote the principal value of the logarithm by the symbol $\text{Ln } z$. Thus, the expression $f(z) = \text{Ln } z$ defines a function, whereas, $F(z) = \ln z$ defines a multiple-valued function.

Definition **Principal Value of the Complex Logarithm**

The complex function $\text{Ln } z$ defined by:

$$\text{Ln } z = \log_e |z| + i\text{Arg}(z) \quad (14)$$

is called the principal value of the complex logarithm.

$$\text{Ln } z = \log_e r + i\theta, \quad -\pi < \theta \leq \pi. \quad (15)$$

EXAMPLE . Principal Value of the Complex Logarithm

Compute the principal value of the complex logarithm $\text{Ln } z$ for

(a) $z = i$

(b) $z = 1 + i$

(c) $z = -2$

Solution In each part we apply (14) of Definition

(a) For $z = i$, we have $|z| = 1$ and $\text{Arg}(z) = \pi/2$, and so:

$$\text{Ln } i = \log_e 1 + \frac{\pi}{2}i.$$

However, since $\log_e 1 = 0$, this simplifies to:

$$\text{Ln } i = \frac{\pi}{2}i.$$

(b) For $z = 1 + i$, we have $|z| = \sqrt{2}$ and $\text{Arg}(z) = \pi/4$, and so:

$$\text{Ln}(1 + i) = \log_e \sqrt{2} + \frac{\pi}{4}i.$$

Because $\log_e \sqrt{2} = \frac{1}{2} \log_e 2$, this can also be written as:

$$\text{Ln}(1 + i) = \frac{1}{2} \log_e 2 + \frac{\pi}{4}i \approx 0.3466 + 0.7854i.$$

(c) For $z = -2$, we have $|z| = 2$ and $\text{Arg}(z) = \pi$, and so:

$$\text{Ln}(-2) = \log_e 2 + \pi i \approx 0.6931 + 3.1416i.$$

Ln z as an Inverse Function

Because $\text{Ln } z$ is *one* of the values of the complex logarithm $\ln z$, it follows from (10) that:

$$e^{\text{Ln } z} = z \text{ for all } z \neq 0. \quad (16)$$

This suggests that the logarithmic function $\text{Ln } z$ is an inverse function of exponential function e^z . Because the complex exponential function is not

$$\begin{aligned} \text{Ln } e^z &= \log_e |e^z| + i \text{Arg}(e^z) \\ &= \log_e e^x + iy \\ &= x + iy. \end{aligned}$$

Thus, we have shown that:

$$\text{Ln } e^z = z \text{ if } -\infty < x < \infty \quad \text{and} \quad -\pi < y \leq \pi. \quad (17)$$

From (16) and (17), we conclude that $\text{Ln } z$ is the inverse function of e^z defined on the fundamental region. The following summarizes the relationship between these functions.

$\text{Ln } z$ as an Inverse Function of e^z

If the complex exponential function $f(z) = e^z$ is defined on the fundamental region $-\infty < x < \infty, -\pi < y \leq \pi$, then f is one-to-one and the inverse function of f is the principal value of the complex logarithm $f^{-1}(z) = \text{Ln } z$.

Analyticity

$$|z| > 0, \quad -\pi < \arg(z) < \pi, \quad (18)$$

$$f_1(z) = \log_e r + i\theta \quad (19)$$

Theorem Analyticity of the Principal Branch of $\ln z$

The principal branch f_1 of the complex logarithm defined by (19) is an analytic function and its derivative is given by:

$$f_1'(z) = \frac{1}{z}. \quad (20)$$

Logarithmic Mapping The complex logarithmic mapping $w = \text{Ln } z$ can be understood in terms of the exponential mapping $w = e^z$ since these functions are inverses of each other. For example, because $w = e^z$ maps the fundamental region $-\infty < x < \infty$, $-\pi < y \leq \pi$, in the z -plane onto the set $|w| > 0$ in the w -plane, it follows that inverse mapping $w = \text{Ln } z$ maps the set $|z| > 0$ in the z -plane onto the region $-\infty < u < \infty$, $-\pi < v \leq \pi$, in the w -plane. Other properties of the exponential mapping can be similarly restated as properties of the logarithmic mapping. The following summarizes some of these properties.

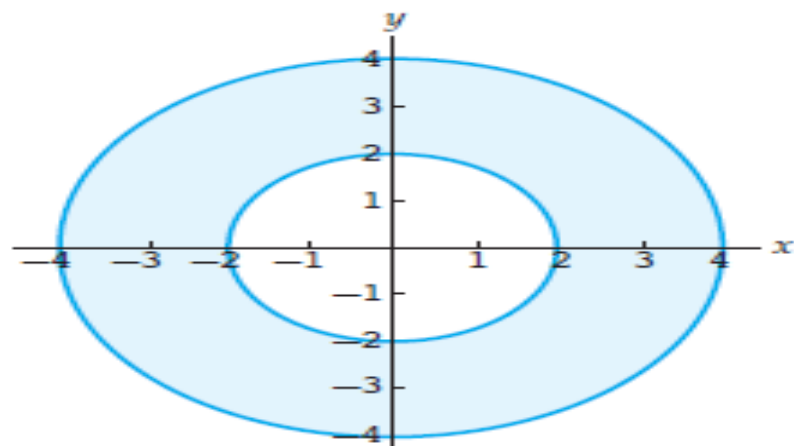
Logarithmic Mapping Properties

- (i) $w = \text{Ln } z$ maps the set $|z| > 0$ onto the region $-\infty < u < \infty$, $-\pi < v \leq \pi$.
- (ii) $w = \text{Ln } z$ maps the circle $|z| = r$ onto the vertical line segment $u = \log_e r$, $-\pi < v \leq \pi$.
- (iii) $w = \text{Ln } z$ maps the ray $\arg(z) = \theta$ onto the horizontal line $v = \theta$, $-\infty < u < \infty$.

EXAMPLE Logarithmic Mapping

Find the image of the annulus $2 \leq |z| \leq 4$ under the logarithmic mapping $w = \text{Ln } z$.

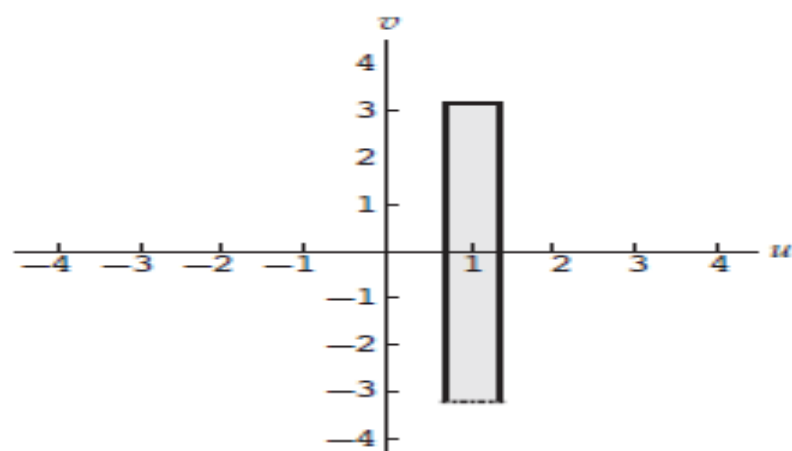
Solution From property (ii) of logarithmic mapping, the boundary circles $|z| = 2$ and $|z| = 4$ of the annulus map onto the vertical line segments $u = \log_e 2$ and $u = \log_e 4$, $-\pi < v \leq \pi$, respectively. In a similar manner, each circle $|z| = r$, $2 \leq r \leq 4$, maps onto a vertical line segment $u = \log_e r$, $-\pi < v \leq \pi$. Since the real logarithmic function is increasing on its domain, it follows that $u = \log_e r$ takes on all values in the interval $\log_e 2 \leq u \leq \log_e 4$ when $2 \leq r \leq 4$. Therefore, the image of the annulus $2 \leq |z| \leq 4$ shown in color in Figure 15.1(a) is the rectangular region $\log_e 2 \leq u \leq \log_e 4$, $-\pi < v \leq \pi$, shown in gray in Figure 15.1(b).



(a) The annulus $2 \leq |z| \leq 4$



$$w = \operatorname{Ln} z$$



(b) The image of the annulus in (a)

Figure The mapping $w = \operatorname{Ln} z$

In each part, determine the image of the set under the given mapping and sketch in the plane.

a) **Sketch** the following region in the complex plane

$$2 < |z| < 4, -\frac{2\pi}{3} < \arg(z) < \frac{2\pi}{3}$$

and

Find the image of the given region under the mapping $\omega = \operatorname{Ln} z$ and sketch it.

b) **Sketch** the following region in the complex plane

$$1 \leq x \leq 2, \frac{\pi}{6} \leq y \leq \frac{\pi}{2}$$

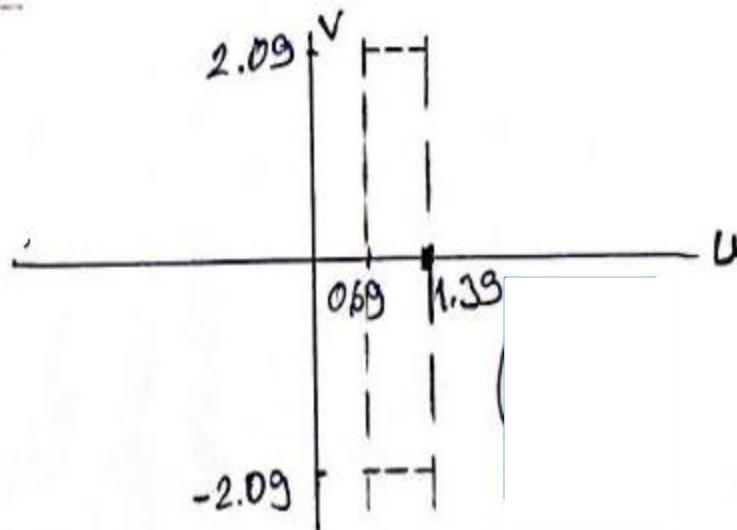
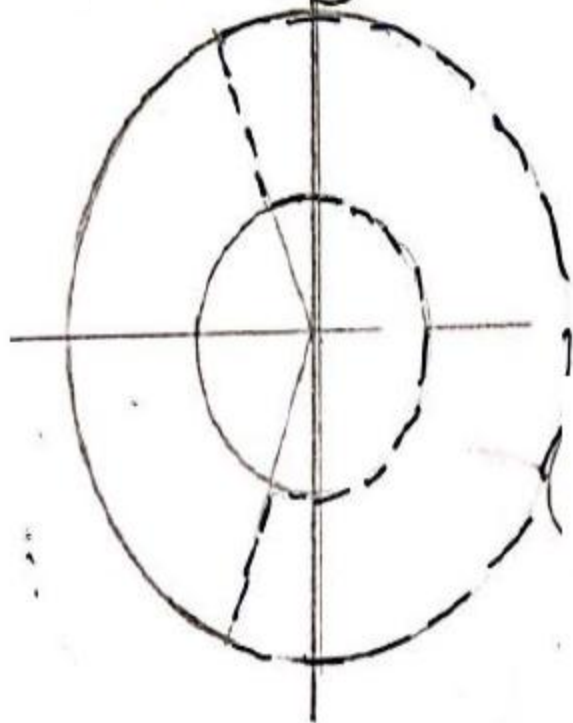
and

Find the image of the given region under the mapping $\omega = e^z$ and sketch it.

$$a) \quad 2 < |z| < 4 \quad -\frac{\pi}{3} < \arg(z) < \frac{\pi}{3} \quad \omega = \ln z$$

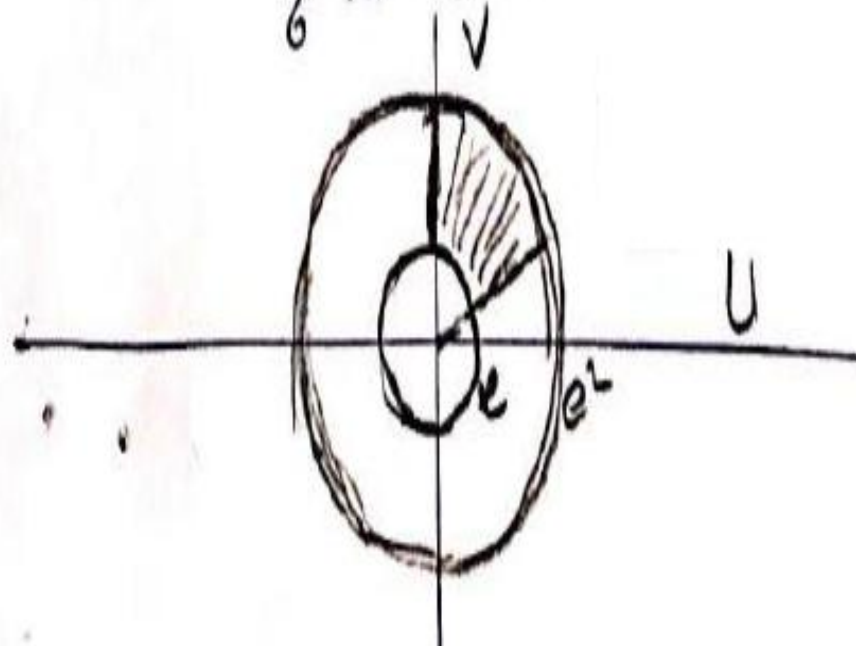
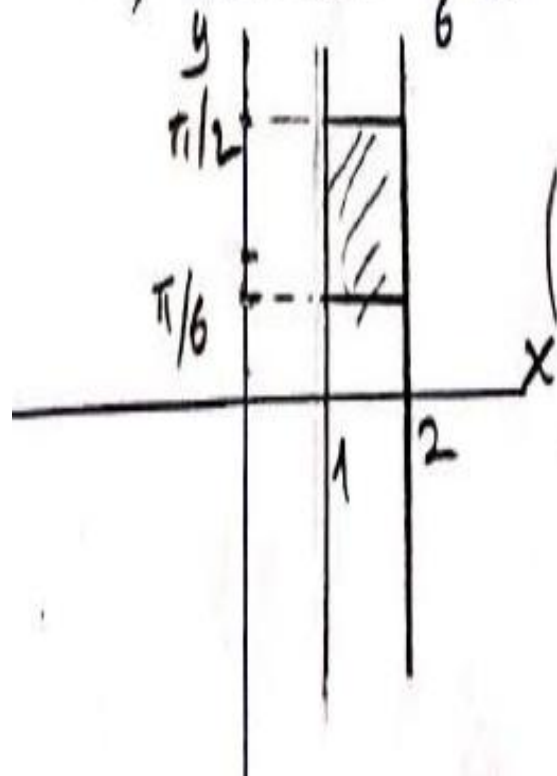
$$\log_e 2 < u < \log_e 4 \Rightarrow 0.69 < u < 1.39$$

$$-\frac{2\pi}{3} < v < \frac{2\pi}{3} \Rightarrow 2.09 < v < -2.09$$



$$b) 1 \leq x \leq 2 \quad \frac{\pi}{6} \leq y \leq \frac{\pi}{2} \quad w = e^z \Rightarrow e^1 \leq |w| \leq e^2$$

$$\frac{\pi}{6} \leq \arg(w) \leq \frac{\pi}{2}$$



Definition 1.1.1 Complex Powers

If α is a complex number and $z \neq 0$, then the **complex power** z^α is defined to be:

$$z^\alpha = e^{\alpha \ln z}. \quad (1)$$

$$z^n = e^{n \ln z} = e^{n[\log_e |z| + i \arg(z)]} = e^{n \log_e |z|} e^{n \arg(z)i}. \quad (2)$$

If $\theta = \text{Arg}(z)$, then $\arg(z) = \theta + 2k\pi$ where k is an integer and so

$$e^{n \arg(z)i} = e^{n(\theta + 2k\pi)i} = e^{n\theta i} e^{2nk\pi i}.$$

we have that $e^{2nk\pi i} = \cos(2nk\pi) + i \sin(2nk\pi)$. Because n and k are integers, it follows that $2nk\pi$ is an even multiple of π , and so $\cos(2nk\pi) = 1$ and $\sin(2nk\pi) = 0$. Consequently, $e^{2nk\pi i} = 1$ and (2) can be rewritten as:

$$z^n = e^{n \log_e |z|} e^{n \text{Arg}(z)i}, \quad (3)$$

which is single-valued.

Although the previous discussion shows that (1) can define a single-valued function, you should bear in mind that, in general,

$$z^\alpha = e^{\alpha \ln z} \quad (4)$$

defines a multiple-valued function. We call the multiple-valued function given by (4) a **complex power function**.

EXAMPLE 1 Complex Powers

Find the values of the given complex power: (a) i^{2i} (b) $(1 + i)^i$.

Solution In each part, the values of z^α are found using (1).

(a) In part (a) of Example 1 we saw that:

You can find it in the 91. slide

$$\ln i = \frac{(4n + 1)\pi}{2}i.$$

Thus, by identifying $z = i$ and $\alpha = 2i$ in (1) we obtain:

$$i^{2i} = e^{2i \ln i} = e^{2i[(4n+1)\pi i/2]} = e^{-(4n+1)\pi}$$

for $n = 0, \pm 1, \pm 2, \dots$. The values of i^{2i} corresponding to, say, $n = -1, 0$, and 1 are 12391.6 , 0.0432 , and 1.507×10^{-7} , respectively.

(b) In part (b) of Example

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$$\ln(1+i) = \frac{1}{2} \log_e 2 + \frac{(8n+1)\pi}{4} i$$

for $n = 0, \pm 1, \pm 2, \dots$. Thus, by identifying $z = 1+i$ and $\alpha = i$ in (1) we obtain:

$$(1+i)^i = e^{i \ln(1+i)} = e^{i[(\log_e 2)/2 + (8n+1)\pi i/4]},$$

or

$$(1+i)^i = e^{-(8n+1)\pi/4 + i(\log_e 2)/2},$$

for $n = 0, \pm 1, \pm 2, \dots$.

Complex powers defined by (1) satisfy the following properties that are analogous to properties of real powers:

$$z^{\alpha_1} z^{\alpha_2} = z^{\alpha_1 + \alpha_2}, \quad \frac{z^{\alpha_1}}{z^{\alpha_2}} = z^{\alpha_1 - \alpha_2}, \quad (5)$$

and $(z^\alpha)^n = z^{n\alpha} \quad \text{for } n = 0, \pm 1, \pm 2, \dots$

Definition **Principal Value of a Complex Power**

If α is a complex number and $z \neq 0$, then the function defined by:

$$z^\alpha = e^{\alpha \text{Ln} z} \quad (6)$$

is called the principal value of the complex power z^α .

EXAMPLE 2 Principal Value of a Complex Power

Find the principal value of each complex power: (a) $(-3)^{i/\pi}$ (b) $(2i)^{1-i}$

Solution In each part, we use (6) to find the principal value of z^α .

(a) For $z = -3$, we have $|z| = 3$ and $\text{Arg}(-3) = \pi$, and so $\text{Ln}(-3) = \log_e 3 + i\pi$. Thus, by identifying $z = -3$ and $\alpha = i/\pi$ in (6), we obtain:

$$(-3)^{i/\pi} = e^{(i/\pi)\text{Ln}(-3)} = e^{(i/\pi)(\log_e 3 + i\pi)},$$

or
$$(-3)^{i/\pi} = e^{-1 + i(\log_e 3)/\pi}.$$

$$e^{-1 + i(\log_e 3)/\pi} = e^{-1} \left(\cos \frac{\log_e 3}{\pi} + i \sin \frac{\log_e 3}{\pi} \right),$$

and so,
$$(-3)^{i/\pi} \approx 0.3456 + 0.1260i.$$

(b) For $z = 2i$, we have $|z| = 2$ and $\text{Arg}(z) = \pi/2$, and so $\text{Ln } 2i = \log_e 2 + i\pi/2$
By identifying $z = 2i$ and $\alpha = 1 - i$ in (6) we obtain:

$$(2i)^{1-i} = e^{(1-i)\text{Ln}2i} = e^{(1-i)(\log_e 2 + i\pi/2)},$$

or
$$(2i)^{1-i} = e^{\log_e 2 + \pi/2 - i(\log_e 2 - \pi/2)}.$$

We approximate this value

$$\begin{aligned}(2i)^{1-i} &= e^{\log_e 2 + \pi/2} \left[\cos \left(\log_e 2 - \frac{\pi}{2} \right) - i \sin \left(\log_e 2 - \frac{\pi}{2} \right) \right] \\ &\approx 6.1474 + 7.4008i.\end{aligned}$$

$$f_1(z) = e^{\alpha(\log_e r + i\theta)}, \quad -\pi < \theta < \pi \quad (7)$$

$$f_1'(z) = \frac{d}{dz} e^{\alpha \text{Ln} z} = e^{\alpha \text{Ln} z} \frac{d}{dz} [\alpha \text{Ln} z] = e^{\alpha \text{Ln} z} \frac{\alpha}{z}. \quad (8)$$

$$\frac{d}{dz} z^\alpha = \alpha z^{\alpha-1}. \quad (9)$$

EXAMPLE 3 Derivative of a Power Function

Find the derivative of the principal value z^i at the point $z = 1 + i$.

Solution Because the point $z = 1 + i$ is in the domain $|z| > 0$, $-\pi < \arg(z) < \pi$, it follows from (9) that:

$$\frac{d}{dz} z^i = i z^{i-1},$$

and so,
$$\left. \frac{d}{dz} z^i \right|_{z=1+i} = i z^{i-1} \Big|_{z=1+i} = i (1 + i)^{i-1}.$$

We can use (5) to rewrite this value as:

$$i (1 + i)^{i-1} = i (1 + i)^i (1 + i)^{-1} = i (1 + i)^i \frac{1}{1 + i} = \frac{1 + i}{2} (1 + i)^i.$$

Moreover, from part (b) of Example 1 with $n = 0$, the principal value of $(1 + i)^i$ is:

$$(1 + i)^i = e^{-\pi/4 + i(\log_e 2)/2},$$

and so
$$\left. \frac{d}{dz} z^i \right|_{z=1+i} = \frac{1 + i}{2} e^{-\pi/4 + i(\log_e 2)/2} \approx 0.1370 + 0.2919i.$$

Show where the function $z \mapsto x^3 + i(1 - y)^3$ is:

- (a) analytic;
- (b) differentiable (here $z = x + iy$).

Solution. If the function

$$f(z) = f(x + iy) = x^3 + i(1 - y)^3$$

were analytic then we will have

$$u_x = v_y \quad \Rightarrow \quad 3x^2 = -3(1 - y)^2,$$

which is only satisfied at $x = 0$, $y = 1$; and, on the other hand,

$$u_y = -v_x \quad \Rightarrow \quad 0 = 0,$$

which holds everywhere. Note also that the components of $f(z)$, and all its first-order partials exist everywhere.

Since the Cauchy-Riemann equations only hold at $z = i$, the function $f(z)$ is only differentiable at $z = i$. Hence, in particular, it is not differentiable on any neighbourhood of any point, and therefore is nowhere analytic.

a) Find the values of $(-1 - \sqrt{3}i)^{1/4}$.

b) Sketch the following region on the complex plane

$$|z - 1 + i| < 2, \text{Arg}(z) > \frac{\pi}{2}$$

a) $(-1 - \sqrt{3}i)^{1/4}$ $| -1 - \sqrt{3}i | = 2$ $\arg(-1 - \sqrt{3}i) = \frac{4\pi}{3} + 2n\pi$
 $n \in \mathbb{Z}$

$$\Rightarrow z_n = 2^{1/4} e^{\frac{\frac{4\pi}{3} + 2n\pi i}{4}} \quad n = 0, 1, 2, 3$$

$$\Rightarrow z_0 = 2^{1/4} e^{\frac{\pi}{3}i} = 2^{1/4} \left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right) = 2^{1/4} \left(\frac{1}{2} + \frac{\sqrt{3}}{2}i \right)$$

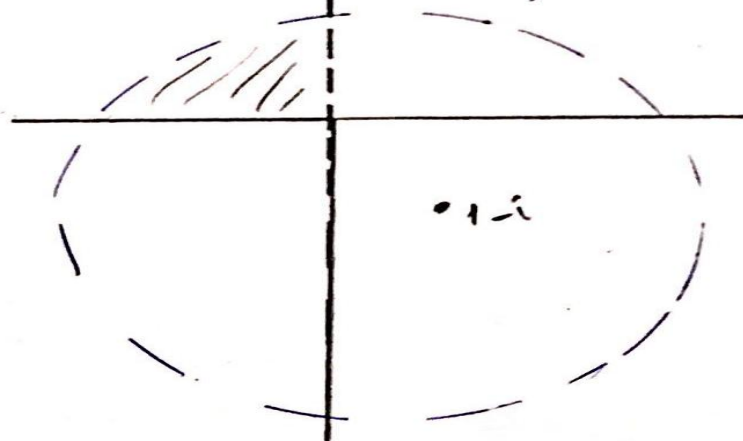
$$z_1 = 2^{1/4} e^{\frac{5\pi}{6}i} = 2^{1/4} \left(\cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6} \right) = 2^{1/4} \left(-\frac{\sqrt{3}}{2} + \frac{1}{2}i \right)$$

$$z_2 = 2^{1/4} e^{\frac{4\pi}{3}i} = 2^{1/4} \left(\cos \frac{4\pi}{3} + i \sin \frac{4\pi}{3} \right) = 2^{1/4} \left(-\frac{1}{2} - \frac{\sqrt{3}}{2}i \right)$$

$$z_3 = 2^{1/4} e^{\frac{11\pi}{6}i} = 2^{1/4} \left(\cos \frac{11\pi}{6} + i \sin \frac{11\pi}{6} \right) = 2^{1/4} \left(\frac{\sqrt{3}}{2} - \frac{1}{2}i \right)$$

b) $|z - 1 + i| < 2$ Center: $1 - i$

$\text{Arg } z > \frac{\pi}{2}$ Vertical cap: 2



a) Find all solutions of $e^{2z-1} = i$.

b) Let $z_1, z_2 \in \mathbb{C}$. Show that $\frac{z_1 - z_2}{2\pi i} \in \mathbb{Z}$ is true, if $e^{z_1} = e^{z_2}$.

1) a) $e^{2z-1} = i \quad z = x + iy$

$$\Rightarrow e^{2(x+iy)-1} = e^{2x-1} e^{2iy} = e^{i\frac{\pi}{2}}$$

$$\Rightarrow x = \frac{1}{2} \quad y = \frac{\pi}{4} + n\pi \Rightarrow \boxed{z = \frac{1}{2} + \frac{\pi i}{4} + n\pi i \quad n=0, \pm 1, \pm 2, \dots}$$

$$b) e^{z_1} = e^{z_2} \Rightarrow \frac{e^{z_1}}{e^{z_2}} = 1 \Rightarrow e^{z_1 - z_2} = 1 = e^{2n\pi i}$$

$$\Rightarrow z_1 - z_2 = 2n\pi i \quad n \in \mathbb{Z}$$

$$\frac{z_1 - z_2}{2\pi i} = \frac{2n\pi i}{2\pi i} = n \in \mathbb{Z} \text{ olur.}$$

Let $u(x, y), v(x, y)$ harmonic functions in region D on the plane. And let v is a harmonic conjugate function of u .

a) Show that $u^2 - v^2$ is a harmonic function in D .

And show that $2uv$ is a harmonic conjugate of $u^2 - v^2$.

b) Find the harmonic conjugate function of $u^3 - 3uv^2$.

$$\nabla^2 u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 \Rightarrow u_{xx} + u_{yy} = 0 \quad \left(\begin{array}{l} u \text{ and } v \\ \text{harmonic fns} \end{array} \right)$$

$$v_{xx} + v_{yy} = 0$$

v is a harmonic conjugate function of u . $\Rightarrow u_x = v_y \quad u_y = -v_x$

a) $\phi = u^2 - v^2 \quad \psi = 2uv$

$$\phi_x = 2uu_x - 2vv_x \quad \phi_{xx} = 2u_x^2 + 2uu_{xx} - 2v_x^2 - 2vv_{xx}$$

$$\phi_y = 2uu_y - 2vv_y \quad \phi_{yy} = 2u_y^2 + 2uu_{yy} - 2v_y^2 - 2vv_{yy}$$

$$\phi_{xx} + \phi_{yy} = 2u(u_{xx} + u_{yy}) - 2v(\underbrace{v_{xx} + v_{yy}}_0) + 2(u_x^2 + u_y^2) - 2(v_x^2 + v_y^2)$$

$$= 2(u_x^2 + u_y^2) - 2(v_x^2 + v_y^2) \quad (u_x = v_y \quad u_y = -v_x)$$

$$= 2(v_y^2 + (-v_x)^2) - 2(v_x^2 + v_y^2) = 0 \Rightarrow \phi = u^2 - v^2$$

is harmonic function

$$\phi = 2uv \quad u_x = v_y \quad u_y = -v_x$$

$$\phi_x = 2u_x v + 2u v_x \quad \phi_y = 2u_y v + 2u v_y$$

$$\phi_x = 2u u_x - 2v v_x \quad \phi_y = 2u u_y - 2v v_y$$

$$\Rightarrow \phi_x = 2v v_y - 2u u_y \Rightarrow \phi_y = -\phi_x$$

$$\phi_y = -2v v_x + 2u u_x \Rightarrow \phi_x = \underline{\phi_y}$$

$\Rightarrow$

$2uv$ is a harmonic conjugate of $u^2 - v^2$

$$b) \quad \psi = u^3 - 3uv^2 \Rightarrow \psi_x = 3u^2u_x - 3u_xv^2 - 6uvu_x$$

$$\psi_y = 3u^2u_y - 3u_yv^2 - 6uvu_y$$

Let $K(x,y)$ is harmonic conjugate function of ψ ,

$$\Rightarrow K_y = \psi_x = 3u^2u_x - 3u_xv^2 - 6uvu_x \quad u_x = v_y \quad u_y = -v_x$$

$$K_y = 3u^2v_y - 3v^2v_y + 6uvu_y \Rightarrow K = 3u^2v - v^3 + g(x)$$

$$-K_x = \psi_y \Rightarrow K_x = -3u^2u_y + 3u_yv^2 + 6uvu_y$$

$$= 3u^2v_x - 3v^2v_x + 6uvu_x$$

$$\Rightarrow 6uvu_xv + 3u^2v_x - 3v^2v_x + \frac{dg}{dx} = 3u^2v_x - 3v^2v_x + 6uvu_x$$

$$\Rightarrow \frac{dg}{dx} = 0 \quad g(x) = A = \text{constant}$$

$$\Rightarrow K(x,y) = 3u^2v - v^3 + A$$

$3u^2v - v^3 + A$ is the harmonic conjugate function of $u^3 - 3uv^2$.

D is a region on the complex plane.

Let $D^* = \{\bar{z} \mid \forall z \in D\}$. If $f(z): D \rightarrow \mathbb{C}$ is a differentiable function, $f^*: D^* \rightarrow \mathbb{C}$, show that $f^*(z) = \overline{f(\bar{z})}$ is differentiable.

$$f(z) = f(x+iy) = u(x,y) + i v(x,y) \quad \text{is differentiable in } D$$

$$u_x = v_y \quad u_y = -v_x \quad \text{Cauchy-Riemann equations are satisfied}$$

$$\begin{aligned} f^*(z) = \overline{f(\bar{z})} &= \overline{f(x-iy)} = \overline{u(x,-y) + i v(x,-y)} \\ &= \tilde{u}(x,y) + i \tilde{v}(x,y) \end{aligned}$$

$$\begin{aligned} \Rightarrow \tilde{u}(x,y) &= u(x,-y) \\ \tilde{v}(x,y) &= -v(x,-y) \end{aligned}$$

$$\Rightarrow \tilde{u}_x = u_x \quad \tilde{u}_y = -u_y \quad \tilde{v}_x = -v_x \quad \tilde{v}_y = v_y$$

$$\begin{aligned} u_x = v_y &\Rightarrow \tilde{u}_x = \tilde{v}_y \\ u_y = -v_x &\Rightarrow -\tilde{u}_y = \tilde{v}_x \end{aligned} \quad \left. \vphantom{\begin{aligned} u_x = v_y \\ u_y = -v_x \end{aligned}} \right) \text{Cauchy-Riemann equations are satisfied}$$

$$\Rightarrow f^*(z) \quad \text{is analytic.}$$